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INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS**

**A
FINAL YEAR PROJECT REPORT
ON**

**POWER QUALITY IMPROVEMENT IN ISOLATED AC MICROGRIDS
USING SPIKING NEURAL NETWORKS FOR SELECTIVE
HARMONIC COMPENSATION**

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Abstract

This project presents the design and simulation of a Spiking Neural Network (SNN)-based Shunt Active Power Filter (SAPF) for selective harmonic compensation in an isolated AC microgrid representing a typical rural electrification system in Nepal. The microgrid consists of a 100 kW Micro-Hydropower Plant (MHP) as the primary generation source, a 50 kW Photovoltaic (PV) system as a supplementary source, a 70 kW linear load, and a nonlinear load that introduces harmonic distortion into the system.

The classical SAPF employs the synchronous reference frame (SRF) theory with Butterworth low pass filters and a PI controlled DC bus to generate reference compensation currents, achieving a Total Harmonic Distortion (THD) of 4.74% compared to 12.86% without compensation. The proposed SNN based controller, trained using simulated data generated from the verified classical model, replaces the reference generation logic while retaining the existing Hysteresis Band Current Controller (HBCC) and power stage. The SNN architecture employs Leaky Integrate-and-Fire (LIF) neurons with surrogate gradient training, achieving a training validation R^2 of 0.995.

Results demonstrate that the SNN based controller successfully compensates up to 7th order harmonics, which represents the dominant harmonic components in the system. The study validates the possibility of applying biologically inspired neural networks to real-time power quality control in isolated microgrids and identifies closed-loop training data distribution as a key research direction for further improvement.

Keywords: Spiking Neural Network, Shunt Active Power Filter, Harmonic Compensation, Total Harmonic Distortion, Isolated Microgrid, Leaky Integrate-and-Fire, Surrogate Gradient Training, Micro-Hydropower, Photovoltaic System, Power Quality

Abbreviations

AC	Alternating Current
ADALINE	Adaptive Linear Neuron
ANN	Artificial Neural Network
AVR	Automatic Voltage Regulator
BPTT	Backpropagation Through Time
CNN	Convolutional Neural Network
CSGNN	Continuous Spiking Graph Neural Network
DC	Direct Current
DSP	Digital Signal Processor/Processing
DSTATCOM	Distribution Static Compensator
ELC	Electrical Load Controller
FFT	Fast Fourier Transform
FPGA	Field Programmable Gate Array
HBCC	Hysteresis Band Current Controller
IGBT	Insulated Gate Bipolar Transistor
IEEE	Institute of Electrical and Electronics Engineers
LIF	Leaky Integrate-and-Fire
LPF	Low Pass Filter
MATLAB	Matrix Laboratory
MHP	Micro-Hydropower Plant
MPPT	Maximum Power Point Tracking
MSE	Mean Squared Error
P&O	Perturb and Observe
PCC	Point of Common Coupling
PLL	Phase-Locked Loop
PSS	Power System Stabilizer
PV	Photovoltaic
PWM	Pulse Width Modulation
R^2	Coefficient of Determination
SAPF	Shunt Active Power Filter
SNN	Spiking Neural Network
SRF	Synchronous Reference Frame
THD	Total Harmonic Distortion
UPQC	Unified Power Quality Conditioner
VSI	Voltage Source Inverter

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1. Introduction

1.1. Background

The global shift toward renewable energy has led to extensive development of microgrids - localized energy systems that can operate independently or in conjunction with the main electrical grid [6]. In the context of Nepal, microgrid technology is not just a technological trend but a socio-economic necessity. Because of the country's rough topography, extending the national central grid to every remote place is both geographically challenging and economically unfeasible. The result has been that isolated AC microgrids, mainly powered by Micro-Hydropower Plants (MHPs), have become the backbone of rural electrification in Nepal [3].

However, as rural electrification continues, the proportion of nonlinear loads such as LED lighting, motor drives, battery chargers and other power electronics-based appliances is steadily increasing. Unlike linear resistive and inductive loads, nonlinear loads draw current in a non-sinusoidal manner, injecting harmonic currents back into the supply network. Because the generator impedance is high and there is no stiff utility grid to absorb disturbances, these harmonics cause measurable degradation like transformer overheating, increased copper and core losses, voltage distortion at the point of common coupling, and interference with sensitive equipment [1]. The IEEE 519-2022 standard limits Total Harmonic Distortion (THD) at the point of common coupling to 5% for most distribution applications which is easily exceeded by unmitigated nonlinear loads on isolated microgrids.

1.2. Problem Statement

The increasing penetration of renewable energy sources and power electronic interfaces in modern AC microgrids has introduced significant power quality challenges, particularly harmonic distortion caused by nonlinear loads such as LED lighting systems, variable speed drives, and rectifier-based appliances [4]. In the context of isolated and rural AC microgrids, where grid stiffness is inherently low and generation capacity is limited, these harmonic currents cause disproportionately severe effects including transformer overheating, increased system losses, voltage distortion, and malfunction of sensitive equipment.

Conventional harmonic mitigation approaches are inadequate in this context. Passive filters are bulky, prone to resonance, offer compensation only at fixed frequencies and poor adaptability to dynamic load conditions. Active power filters are more flexible, but their effectiveness depends critically on the speed and accuracy of harmonic detection, and the responsiveness of the current control strategy used to drive the compensating inverter. Most existing harmonic detection techniques rely on frequency-domain transformations such as the Fast Fourier Transform (FFT) or the instantaneous reactive power (pq) theory, both of which introduce processing latency and are computationally demanding for embedded controllers. Fixed-band Hysteresis Band Current Controllers (HBCCs), which are commonly used to drive the compensating inverter, operate with a constant switching tolerance that cannot respond to the varying severity of harmonic distortion under changing load conditions. Rule-based adaptive controllers offer partial improvement but depend on predefined heuristics that cannot generalize across the wide range of operating conditions encountered in practice.

While conventional Artificial Neural Networks (ANNs) have been successfully applied to SAPF harmonic extraction, they process inputs as continuous floating points activations at every timestep. This imposes a constant computational energy cost and requires explicit recurrent structures to capture temporal signal dynamics. When it comes to designing a system in which the control electronics have to work with very limited power budgets, harmonic content will be inherently time-dependent, and

embedding onto an electronic board located at rural environment will be required, then the concept of a sparse, event-based, time-aware computation appears highly attractive.

1.3. Proposed Approach: SNN-Based Selective Harmonic Compensation

This project proposes the use of a Spiking Neural Network (SNN) as the core harmonic detection and reference current generation mechanism within a Shunt Active Power Filter (SAPF). Spiking Neural Networks are a neuromorphic computing paradigm in which information is encoded and processed through discrete spike events rather than continuous activations [25]. Their event-driven nature makes them inherently efficient for processing non-stationary time-domain signals, and their ability to operate directly on raw waveform data eliminates the need for frequency-domain preprocessing.

The trained SNN takes instantaneous three-phase voltage at Point of Common Coupling (PCC), instantaneous load current measurements, DC busbar voltage and actual current injected by VSI as inputs and directly outputs three-phase reference compensating currents. These references are tracked by a fixed-band HBCC that drives the IGBT switches of the VSI, injecting the compensating currents into the AC bus to compensate for the harmonic content introduced by the nonlinear load. The PV system simultaneously supplies real power to the microgrid, making the compensation fully renewable-powered with no additional energy source required.

The proposed system is implemented and evaluated entirely in MATLAB/Simulink and compared against a classical dq-theory based SAPF operating under identical conditions [12]. The primary performance metric is the reduction of source current THD to below 5% in compliance with IEEE 519-2022, alongside assessment of reactive power compensation, DC bus stability, and current tracking accuracy.

1.4. Objectives

- To model an isolated AC microgrid in MATLAB/Simulink featuring a 100 kW micro-hydro plant and a 50 kW PV system integrated with non-linear loads, with an Electronic Load Controller (ELC) for frequency regulation.
- To implement and validate a classical Shunt Active Power Filter (SAPF) using dq-theory based reference current generation and a fixed-band Hysteresis Band Current Controller as a performance baseline for comparison.
- To design and train a Spiking Neural Network using Leaky Integrate-and-Fire (LIF) neurons and surrogate gradient methods for time-domain classification of harmonic severity from raw voltage and current signals at the Point of Common Coupling (PCC).
- To integrate the trained SNN into the SAPF control loop as a replacement for the classical reference generation mechanism, enabling selective harmonic compensation without reliance on frequency-domain transformations.
- To evaluate the system performance in compliance with the IEEE 519 standard to meet the Total Harmonic Distortion (THD) requirement of less than 5%.
- To perform a comparative analysis between the classical and proposed SNN-based compensation system in terms of harmonic mitigation and switching efficiency.

1.5. Scope

The scope of this project is centered on the replacement of traditional mathematical reference current generation techniques with a neuromorphic computing approach. It involves the development of an SNN-based module within a Shunt Active Power Filter (SAPF) to handle the extraction of harmonic currents from an AC microgrid. The system under study comprises a three-phase isolated AC microgrid consisting of a 100 kVA micro-hydro synchronous generator and a 50 kW photovoltaic system, supplying a nonlinear load emulated by a three-phase diode bridge rectifier with an RL filter. The compensation system is implemented as a Shunt Active Power Filter, which handles harmonic compensation. Two control configurations are modeled and compared: a classical dq-theory based SAPF with fixed-band HBCC, and a proposed SNN-based SAPF with fixed-band HBCC.

The SNN is trained offline on synthetic harmonic waveform data generated from the Simulink model and its trained weights are imported into the simulation. The study is conducted entirely under controlled simulation conditions at fixed irradiance and temperature (Standard Test Conditions). Performance is assessed against IEEE 519-2022 standards using THD, harmonic spectrum, DC bus stability, and current tracking accuracy as evaluation metrics.

The scope is limited to steady-state and dynamic simulation under controlled irradiance and temperature conditions and the study focuses on current harmonic mitigation. Grid synchronization, islanding detection, protection coordination, and energy storage integration are outside the scope of this work. The results are interpreted with reference to the IEEE 519 standard THD limit of 5% for current harmonics.

1.6. Limitations

A primary limitation of this research is its reliance on a simulated environment. While MATLAB/Simulink provides a robust platform for testing power electronics and neural networks, it may not capture all the physical nuances of a hardware implementation, such as high-frequency noise, sensing delays, and the thermal limitations of physical FPGAs (Field-Programmable Gate Arrays) or neuromorphic chips. On top of it, the SNN module's performance is highly dependent on the quality and diversity of the training data; its ability to generalize to harmonic signatures or load types not included in the training set represents a potential constraint. Additionally, the project focuses specifically on Shunt Active Power Filters and harmonic compensation at the Point of Common Coupling (PCC), meaning it does not address other power quality issues such as deep voltage sags or long-duration outages that would require significant energy storage solutions beyond the scope of an SAPF. Also, the nonlinear load is represented by a single three-phase diode bridge rectifier with a fixed RL filter. Load diversity, step changes between multiple load types, and unbalanced loading conditions are not fully explored.

2. Literature Review

2.1. Background

Harmonic distortion has emerged as one of the most pressing power quality concerns in both modern distribution networks and isolated rural microgrids, mainly due to the increasing trend of power-electronic loads. Bollen [1] provided the theoretical background necessary for investigating the parameters of harmonics, voltage sags, and the operating limits of shunt compensation devices. Ghosh and Ledwich [2] extended the analysis to a device level and studied the role of shunt-connected custom power devices such as DSTATCOM and UPQC in mitigating the reactive power and harmonic issues in the distribution networks. In the context of Nepal where the community-operated Micro-Hydropower Plants (MHPs) more than 3,000 serve remote villages as the supply source of electricity, the problem is more difficult due the limited impedance of small synchronous generators which magnifies the harmonic voltages at the Point of Common Coupling (PCC) creating insulation breakdown, motor overheating, and relay misoperation [3][4][5]. The comprehensive reviews clearly state that the active compensation is the most suitable mitigation technique for the isolated microgrid, although practical implementation in the poor rural area is still an unresolved issue [6].

Classical SAPFs based on Synchronous Reference Frame (SRF) theory offer reliable harmonic attenuation, but their dependence on Low Pass Filters (LPFs) for fundamental extraction introduces inherent phase delay and reduces adaptability in scenarios featuring rapidly varying, stochastic load conditions [11][12]. Spiking Neural Networks (SNNs), identified by Maass [25] as the third generation of neural network models, present a biologically-inspired, time-domain alternative in which information is carried by discrete spike events rather than continuous activations, an approach that inherently suits non-stationary power waveform processing [26][27]. This review covers four interconnected domains: (1) harmonic distortion in isolated MHP microgrids, (2) classical and AI-enhanced SAPF control, (3) hysteresis-band current control for VSI switching, and (4) SNN theory, training, and implementation, highlighting research gaps that inspire the current work.

2.2. Harmonic Distortion in Isolated Microgrids and Rural Electrification Context

2.2.1. Power Quality Fundamentals and the Isolated Microgrid Problem

Bollen [1] pioneered the treatment of power quality degradation metrics (voltage sags, interruptions and harmonic indices) which is still the reference source for power quality degradation metrics for distribution systems. Ghosh and Ledwich [2] are the closest in literature to our work and provide a device-level analysis of shunt compensators operating on weak distribution buses. This is especially relevant in the isolated MHP architecture in which a single synchronous machine provides the only source of voltage to all loads with no stiff utility grid to absorb any disturbances. In islanded operation, harmonic currents injected by the nonlinear loads are reflected by the finite synchronous reactance of the generator and cause harmonic voltage drops all over the PCC waveform, which is the power quality problem identified by Khan, Vijay, and Doolla [6] as the most prevalent problem in isolated microgrids. Their state-of-the-art survey recommends active filtering with a controller capable of handling a lack of voltage reference and generator frequency excursions for islanded systems. In a three-phase system with six-pulse rectifier loads (similar to many rural electrification systems), the 5th and 7th harmonics are dominant and the ELC that maintains constant loading on the generator in an MHP system introduces switching harmonics.

2.2.2. Nepal's MHP Microgrids: Scale, Context, and Power-Quality Challenges

The hilly and mountainous terrain of Nepal poses significant logistical and economic barriers to the grid extending to remote communities, and isolated MHP systems are the mainstay of rural electrification. Mallik et al. [3] documented the successful commissioning of over 3,000 standalone MHP projects, totaling roughly 25 MW, to replace kerosene lamps and diesel generators in thousands of villages across the country in 2019, but Shrestha et al. [4] highlight power quality deterioration, characterized by high THD, protection mis-coordination, and frequent equipment failure, as the main operational threat to the long-term viability of such systems. The widespread adoption of LED drivers, variable speed irrigation pumps and agro-processing rectifiers have fundamentally changed the usually resistive load profile to highly non-sinusoidal, generating harmonic currents that original system designs were not built to handle. Karki et al. [5] phrases this in a socio-economic context, and shows via two case studies in rural Nepal that harmonic-induced equipment damage and voltage instability poses the greatest risk to loads that can be economically justified for rural electrification, i.e., productive-use loads, thereby confirming the theoretical need for active harmonic compensation in islanded MHP systems.

2.2.3. Harmonic Standards: IEEE 519

IEEE Standard 519-2022 [7] defines current total harmonic distortion at the point of common connection (PCC) to a maximum of 5% for system generating converters with a short-circuit ratio greater than 20. In a dynamic microgrid environment with fluctuating renewable generation and a distorted PCC voltage, this standard demands active, real-time compensation capable of constantly adjusting to the instantaneous harmonic content of the load current. Though, passive filters are widely used in rural settings for their simplicity and low cost, they only compensate at their resonant frequency and can be prone to resonance with the generator output impedance, thus making them unsuitable as a stand-alone solution in the ever-changing nonlinear load landscape described in references [3][4][5].

2.3. Shunt Active Power Filters: Classical Control Strategies

2.3.1. Foundational Active Filter Theory and Technology Overview

The theoretical basis of SAPF control was transformed by Akagi, Kanazawa, and Nabae [8], who introduced the Instantaneous Reactive Power (p-q) theory in 1984. They proposed the Instantaneous Reactive Power (p-q) theory based on the Clarke transformation from three-phase voltage and current to the stationary $\alpha\beta$ frame that allowed a simple definition of instantaneous active and reactive power in time domain and enabled energy-storage-free shunt compensation. Akagi reviewed the installed real-world records for the first time [9] and outlined the multifunctional compensator in three key functions: harmonic reduction, reactive power compensation, and voltage regulation control, which is practically the scenario of this project. Singh, Al-Haddad, and Chandra [10] later provided the definitive classification and design guidelines for all the three- and four-wire filter topologies with different control strategies and selection rules, especially the comparison of hysteresis, ramp comparison, and predictive current controllers. They also clarified the drawbacks of the proposed SRF-HBCC and suggested the next architecture, which is the current focus of this project.

2.3.2. Synchronous Reference Frame (dq) Control: Performance and Limitations

The SRF method uses the Park transformation to convert the $\alpha\beta$ frame into a dq frame rotating synchronously with the fundamental frequency, tracked by a Phase-Locked Loop (PLL). In this rotating frame, the 50 Hz fundamental appears as a DC quantity while harmonic components manifest as AC ripple, allowing their separation via LPFs. Al-Gahtani and Al-Ghamdi [11] analyzed an enhanced SRF-based SAPF to protect power transformers and found that although the classical dq method provides good harmonic mitigation in a balanced system, it suffers from poor speed due to LPF phase filtering. The improvement involved adding positive sequence detection and negative sequence detection to improve the speed. However, one problem still persists in their architecture due to fundamental bandwidth limitation associated with harmonic extraction through the use of LPF filter, which causes latency to increase when higher harmonics are involved. Srinivas and Rao [12] conducted a simulation study of SRF-SAPF using a three-level diode clamped inverter, which indicated that there was a significant improvement in THD from around 27% to below 4% due to balanced non-linear loading, similar to the baseline used in this project. Murugesan et al. [13] proved that the inclusion of an adaptive neural network on a DSP platform could result in faster settling and reduced THD.

2.4. AI-Enhanced and Neural Network-Based SAPF Control

2.4.1. ANN-Based Harmonic Extraction and Reference Generation

The use of ANNs for SAPF reference current generation has been studied progressively over the past two decades. According to the findings by Bhattacharya and Chakraborty [14], ADALINE networks are capable of extracting the fundamental frequency components directly from distorted load currents through the implementation of an adaptive weight update algorithm that estimates the values of the fundamental coefficients in the time domain. The process does away with coordinate transformation of the currents, proving that learned techniques have the potential of estimating time-domain references for harmonics accurately – a principle that forms the core of the proposed SNN technique. Singh et al. [15] extended this with dual-axis ADALINE networks to estimate d- and q-components, and observed better performance than that of the traditional LPF in terms of distortion immunity where the input voltage itself has been distorted, a phenomenon common in islanded MHP microgrids. Nascimento et al. [16] continued the trend with the employment of deep ANN-based techniques for the direct time-to-time learning of waveforms in harmonic estimation applications within smart grid active filters. Their experimental validation on real load data establishes the methodological precedent for the simulation-driven SNN training pipeline used in the present work.

2.4.2. Spiking and Hybrid AI Approaches for SAPF

The potential applications of spiking computing for power quality tasks are yet another emerging area of study. Prabhu et al. [17] used Orangutan Optimization Algorithm in combination with Continuous Spiking Graph Neural Networks (CSGNNs) as an approach to active power filter control. They claimed that a THD level of 0.11% was achieved, which exceeded that obtained using ANN, CNN, and genetic algorithms as well as becoming the first time spiking computation was applied to such a task. In their work, the CSGNN served as a module for the prediction stage inside the overall optimization algorithm loop rather than providing real-time references as needed. The present project advances this concept by deploying a standard LIF-SNN as a drop-in replacement for the dq-LPF reference generation block, operating without an optimization loop and enabling low-latency online inference in an isolated microgrid. Rahman et al. [18] came up independently to another similar idea

showing that dividing the whole process into two modules for the reference generator and the DC-link respectively provided better dynamics than a single-stage solution.

2.4.3. PV-Integrated SAPF Systems

The utilization of PV DC link for simultaneous power feed of SAPF inverter and compensation current injection has received much attention for its sustainability in mitigating harmonics. Benaissa et al. [19] analyzed the PV-driven hybrid APF employing three-level neutral-point-clamped converter with P&O MPPT and showed that the PV generation system can keep the THD level at below 5% even with passing clouds if the DC link PI control maintains sufficient voltage margin. The above findings were considered to justify the selection of a 50 kW PV generation system in this study and the critical point of DC bus voltage regulation in identifying valid data ranges for SNN training. According to Patel and Kumar [20], the time lag associated with P&O MPPT operation during irradiance step changes, usually within 3 to 5 seconds, significantly influences the performance of SAPF inverter startup and that the compensation data collected during this transitional stage cannot be considered reliable.

2.5. Hysteresis-Band Current Control for VSI Switching

2.5.1. Classical Fixed-Band HBCC and Its Role in SAPF

One of the most common types of current controller used for SAPFs is the fixed-band Hysteresis-Band Current Controller (HBCC), which is appreciated due to its relative simplicity, stable operation, and quick reaction. The basic principle of its work lies in detecting an error between the measured value of inverter current and its reference one within a fixed band of $\pm HB$ and switching the relevant IGBT transistors when this error goes out of this limit. According to the results reported by Naidu and Singh [21], a detailed investigation of shunt APF operating under dq control and with a fixed HBCC in different nonlinear loads allowed estimating the primary design compromise in terms of its parameters: the more narrow the width of this band, the better THD performance; however, at the same time, the switching frequency would become higher and losses more significant. Their analysis establishes that the band width must be selected to balance these competing objectives for the anticipated range of harmonic severity, the approach adopted in this project, where a fixed band is selected based on worst-case loading conditions.

2.5.2. Harmonic Detection as the Key Determinant of SAPF Performance

Asiminoaei et al. [22] presented a thorough analysis of the effects of different harmonic detection schemes for applications in SAPFs through a decoupled simulation approach, which separates the detection process from the power process. The main contribution of their study is that the selection of the detection scheme, and not the type of inverter nor the switching scheme plays the decisive role in defining the global SAPF performance. LPF-based dq detection causes systematic phase lag, which increases with increasing order of the harmonics and also becomes more severe in case of non-stationary loads. In light of this result, one can find solid justification for the investigation of a learning-based time domain detection method, like the SNN used in this research.

2.5.3. Switching Frequency Behaviour of Fixed-Band HBCC

A known drawback of fixed-band HBCC is that its switching frequency varies with the rate of change of the reference current, the DC bus voltage, and the instantaneous supply voltage, all of which fluctuate during normal microgrid operation. Szromba [23] explored the impact of this feature in a SAPF setting and suggested using stabilized-frequency hysteresis control to prevent the introduction of high-frequency harmonics in the network, explaining that proper sizing of the coupling inductor and passive filtering of the output current are essential in mitigating the ripple current generated by the process. These were important factors considered in designing the coupling inductor and output filter in the simulation setup used in this project. On another note, Mahapatra et al. [24] showed that an event-driven SNN could accurately categorize transient and steady-state power system disturbances from Phasor Measurement Unit (PMU) time-domain data in real-time, providing a concrete example of how SNNs can be applied in processing distorted PCC waveforms in a harmonic compensation scenario.

2.6. Spiking Neural Networks: Theory, Training, and Implementation

2.6.1. The SNN Paradigm: Biological Inspiration and Computational Foundations

Maass [25] formally classified SNNs as the third generation of neural network models and established their theoretical computational universality. Unlike the first generation of perceptrons and the second generation of rate-coded artificial neural networks (ANNs), SNNs represent information in terms of the precise timing and frequency of discrete binary spikes that mimic the electrochemical interactions of biological neurons. Tavanaei et al. [26] review various deep learning techniques for SNNs such as spike-timing dependent plasticity (STDP), ANN to SNN transfer learning, and end-to-end SNN training using backpropagation through time and demonstrate energy and latency benefits compared to ANNs when processing temporal data, which is a motivating factor for adopting SNNs to generate time-domain harmonics references. Roy et al. [27] extend the scope of this work in a comprehensive review of neuromorphic computing hardware ranging from Intel's Loihi to IBM's TrueNorth and prove that spike-based, event-driven inference is orders of magnitude energy efficient compared to traditional deep learning. Such algorithm-hardware co-design considerations further justify the use of an SNN-based harmonic controller over a DSP-based ANN because of energy savings, especially when considering that the control electronics in rural Nepal have stringent power constraints.

2.6.2. The LIF Neuron Model and FPGA Implementation

The Leaky Integrate-and-Fire (LIF) model is the most commonly utilized spiking neuron model in practice. The LIF neuron's membrane voltage integrates the weighted synaptic inputs and then exponentially decreases towards its resting level. Once the threshold level is exceeded, a spike is fired and the voltage is reset. Gao et al. [28] developed an FPGA multi-compartment LIF-based neuromorphic learning chip which demonstrated that the discretized nature of LIF neurons results in efficient mappings to digital circuitry, leading to higher resource usage efficiency and power efficiency compared to previous on-chip implementations - a feasibility conclusion directly applicable to the deployment of the proposed SNN harmonic compensator on an embedded FPGA controller. Ali, Navardi, and Mohsenin [33] developed a first-order LIF SNN on a Xilinx Artix-7 FPGA, the same FPGA family used by embedded power electronics controllers, achieved an impressive 86% improvement in energy efficiency relative to a CNN baseline, proving the effectiveness of the spike communication scheme in reducing energy consumption on an FPGA.

2.6.3. Surrogate Gradient Training

Training SNNs via backpropagation is complicated by the non-differentiability of the Heaviside spike-generation function, which yields zero gradient almost everywhere. Neftci, Mostafa, and Zenke [29] resolved this by introducing surrogate gradient learning, which replaces the spike function's true derivative in the backward pass with a smooth surrogate function, effectively enabling gradient-based optimization of spiking networks and bringing the power of modern deep learning to neuromorphic systems. Zenke and Ganguli [31] demonstrated the approach with the SuperSpike algorithm, proving that multilayer SNNs can be trained under supervised learning by approximating loss gradients, unlocking high precision temporal pattern recognition. In this project, a sigmoid surrogate is used, maintaining the hard threshold in the forward pass for physical correctness while using the sigmoid derivative in the backward pass to allow stable gradient flow.

2.6.4. SNNs for Time-Series Processing and Energy Efficiency

The temporal processing capability of SNNs, rooted in the LIF neuron's inherent memory of past inputs through its membrane potential, makes them naturally suited for time-series applications such as power signal harmonic analysis. Abillama et al. [32] demonstrate that one-hot multi-level LIF SNNs achieve 20 times lower energy consumption than equivalent ANN architectures while maintaining comparable accuracy, and their energy-per-inference analysis is directly applicable to the argument that SNN-based harmonic controllers deployed on neuromorphic chips would significantly reduce the computational power overhead of the SAPF control system. For the 50 Hz power waveform sampled at 100 kHz in this project, highly structured temporal sequences are produced where harmonic components at 250 Hz (5th) and 350 Hz (7th) appear as periodic patterns that the LIF membrane dynamics can integrate and distinguish without requiring explicit recurrent connections.

2.7. Identified Research Gaps and Positioning of the Present Work

The foregoing review reveals a well-established body of knowledge in each constituent domain: rural MHP microgrids [1 - 6], classical SAPF control [7 - 13], ANN-based and spiking harmonic extraction [14 - 18], PV-fed SAPF [19][20], HBCC and harmonic detection [21 - 24], and SNN theory and hardware [25 - 33]; yet no existing work integrates all these elements into a unified system. Four research gaps are precisely identified:

Gap 1: SNN for real-time harmonic reference generation:

While Prabhu et al. [17] demonstrate spiking computation for SAPF optimization, no work has implemented a standard LIF-SNN as a direct, online replacement for the dq-LPF reference generation block in a three-phase SAPF, operating in the time domain without coordinate transformations.

Gap 2: Isolated MHP microgrid context:

All ANN and SNN-enhanced SAPF studies are conducted in grid-connected or large-generator contexts. The unique challenges of an isolated synchronous generator: frequency drift, ELC ballast dynamics, finite fault impedance, and absence of an infinite bus, have not been addressed in AI-based SAPF design.

Gap 3: PV-fed SAPF with SNN in an isolated microgrid:

PV-integrated SAPF studies [19][20] employ classical SRF control. No work combines PV-sourced DC link power, LIF-SNN harmonic extraction, and fixed HBCC in a single islanded microgrid architecture.

Gap 4: Selective harmonic targeting (up to 7th):

Most SAPF studies target broad-spectrum harmonic mitigation. SNN architectures specifically trained for selective extraction up to 7th harmonics, computationally efficient for resource-constrained rural microgrids, have not been studied.

The present project addresses all four gaps by proposing and simulating a neuromorphic harmonic compensation system in which an LIF-SNN replaces the dq-LPF reference generation block, trained on data from an isolated 100 kW MHP integrated with 50 kW PV microgrid in MATLAB/Simulink, and coupled to a fixed-band HBCC. This represents the first integration of neuromorphic computing with isolated-microgrid SAPF design tailored for the Nepali rural electrification context.

3. Methodology and Related Theory

This section describes the theoretical foundations, system configuration, and implementation methodology of the proposed system. The proposed system integrates an isolated AC microgrid with a photovoltaic-fed Shunt Active Power Filter (SAPF) whose reference generation is performed by a Spiking Neural Network (SNN) coupled to a fixed Hysteresis Band Current Controller (HBCC). Each component of the system is described in terms of both its governing theory and its specific implementation in MATLAB/Simulink.

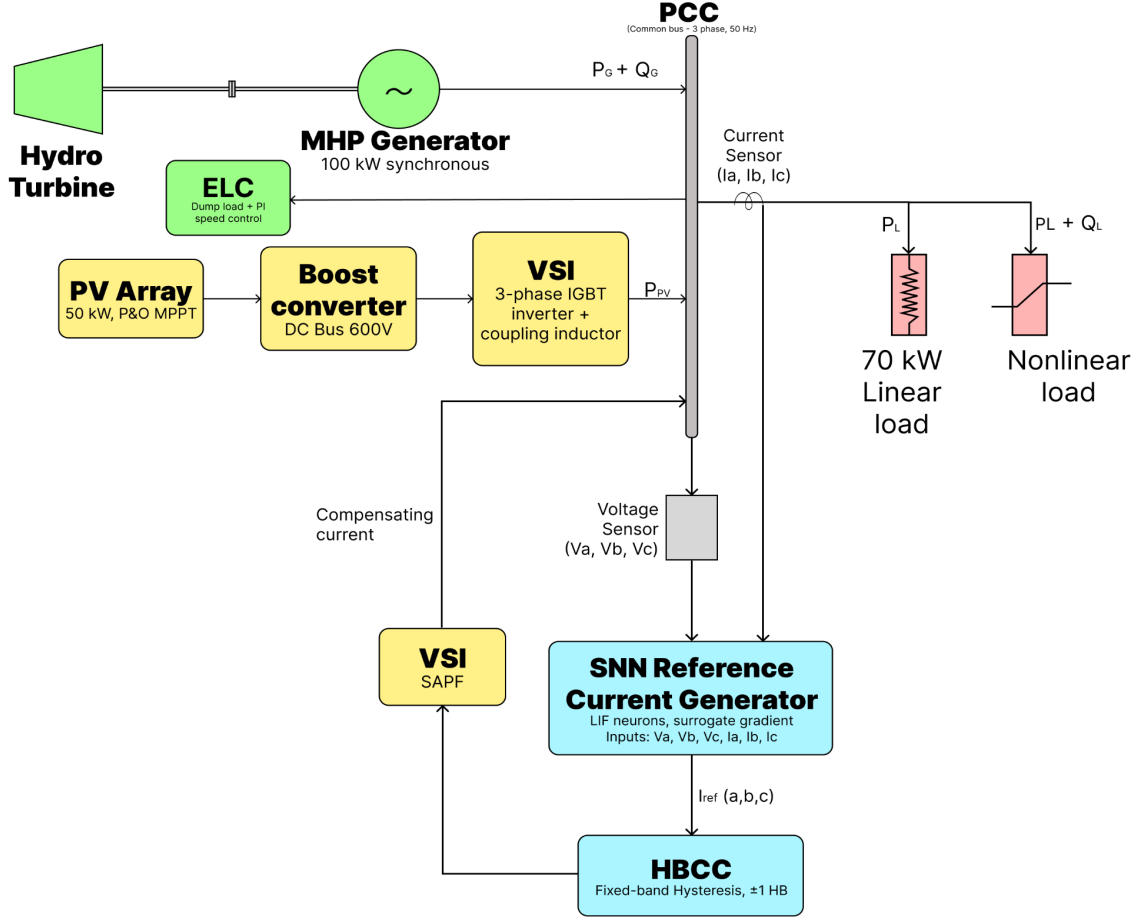


Figure 1: Block Diagram of Proposed System

3.1. System Overview

The system under study is a three-phase isolated AC microgrid supplied by two sources: a micro-hydro synchronous generator and a photovoltaic (PV) system. Both sources deliver power through a common AC busbar to a mixed load comprising a balanced three-phase linear load and a nonlinear load modelled by a three-phase diode bridge rectifier with an RL filter. The nonlinear load injects harmonic currents into the network, degrading power quality at the Point of Common Coupling (PCC). Frequency regulation is handled by an Electronic Load Controller (ELC) connected to the micro-hydro generator, which diverts excess generation to a dump load whenever consumer demand falls.

The VSI transfers PV-generated real power to the AC grid. Harmonic mitigation is achieved by a Shunt Active Power Filter (SAPF). Two control configurations of this SAPF are implemented and compared: a classical SRF dq-theory based controller serving as the performance baseline, and the proposed SNN-based controller as the primary contribution.

Component	Rating / Type	Key Parameters
Micro-Hydropower Plant	100 kW, Synchronous Generator	AVR (Excitation System) + ELC, 3-phase AC output
Photovoltaic System	50 kW, Grid-forming VSI	MPPT (P&O), Boost Converter, Stabilizes at t = 3s
Electronic Load Controller	Dump Load Switching	PI control via ω_m feedback
Shunt Active Power Filter	VSI + Coupling Inductor	DC Bus 600V, HBCC switching
Linear Load	70 kW, Resistive-Inductive	Balanced three-phase
Nonlinear Load	Rectifier-based ($R = 5 \Omega$, $C = 1000 \mu F$)	Source of 5th and 7th harmonics
Simulation Solver	Discrete Fixed-Step	1e-5 s (100 kHz sampling)

Table 1: System Overview

3.2. Microhydro Generator Model

The micro-hydro generation unit is represented as a three-phase synchronous machine operating in generator mode using the standard per-unit (pu) representation available in Simscape Electrical, which accounts for both the electrical and mechanical dynamics of the machine.

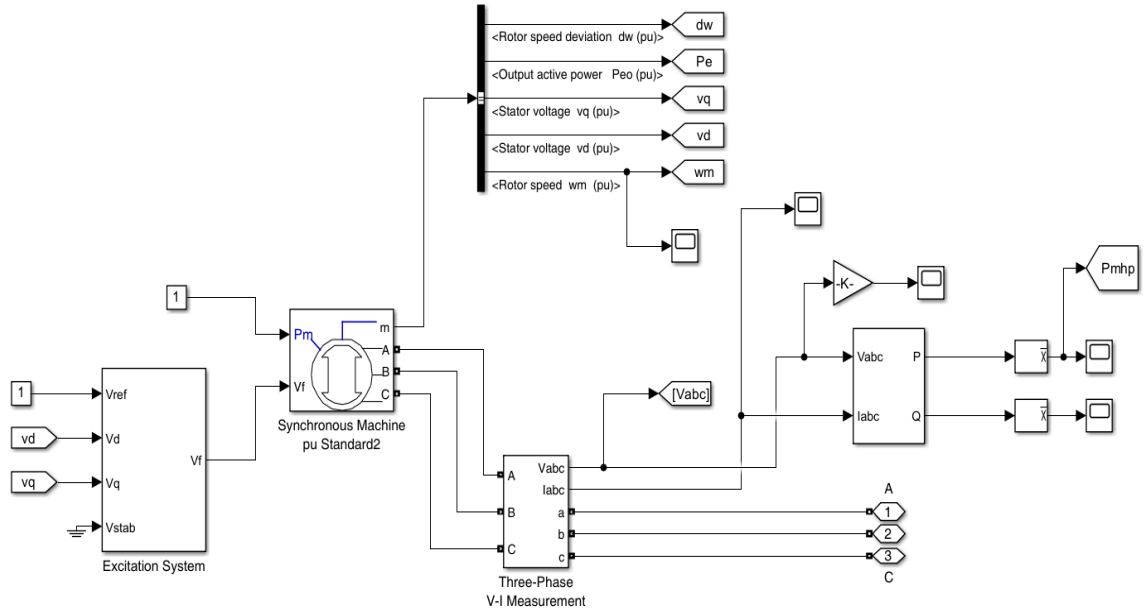


Figure 2: 100 kW Micro Hydropower Plant Model in MATLAB/Simulink

3.2.1. Synchronous Machine Equations

The electrical behavior of the synchronous machine in the dq0 reference frame is governed by the following voltage equations:

$$\begin{aligned}
 v_d &= -R_s * i_d + \frac{d}{dt}(\lambda_d) - \omega_r \lambda_q \\
 v_q &= -R_s * i_q + \frac{d}{dt}(\lambda_q) + \omega_r \lambda_d
 \end{aligned}$$

where v_d and v_q are the d-axis and q-axis stator voltages, R_s is the stator resistance, i_d and i_q are the stator currents, λ_d and λ_q are the flux linkages, and ω_r is the rotor angular speed.

$$T_e = \frac{3}{2} * \frac{P}{2} * (\lambda_d * i_g - \lambda_g * i_d)$$

$$J * \frac{d}{dt}(\omega_r) = T_m - T_e = B\omega_r$$

In the simulation, the mechanical input is held constant at 1 pu representing steady hydraulic torque.

Terminal voltage is regulated by an Automatic Voltage Regulator (AVR) connected to the field winding of the synchronous machine. The excitation system receives the measured d-axis and q-axis voltages as feedback signals along with a reference voltage ($V_{ref} = 1$ pu), and produces the field voltage (V_f) required to maintain the terminal voltage at the rated level. The stabilizing input (V_{stab}) is set to zero in the absence of a Power System Stabilizer (PSS).

In an isolated micro-hydro system, the generator speed and hence the system frequency is determined by the balance between mechanical input power and electrical load. Since hydraulic turbines cannot rapidly throttle their input, frequency is regulated by maintaining a constant total electrical load through an Electronic Load Controller (ELC).

$$D(t) = K_n * e(t) + K_i * \int e(t)dt$$

Dump load power (P_{dump}) is measured and logged as an indicator of generation surplus.

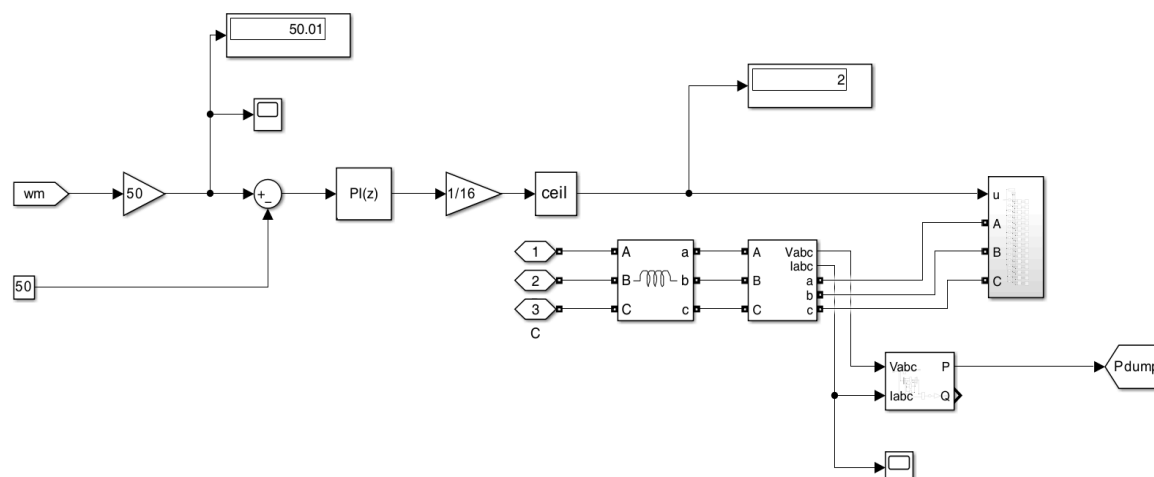


Figure 3: ELC Block in MATLAB/Simulink

There are 16 individual loads each of 5kW integrated together to run as a 80kW ELC.

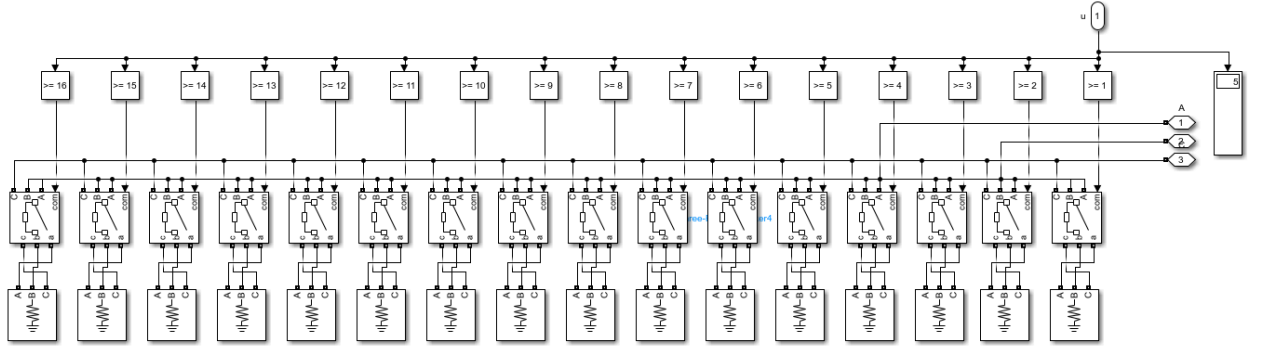


Figure 4: ELC Ballast Load Block in MATLAB/Simulink

3.4. Photovoltaic System

The photovoltaic generation unit comprises a PV array, a DC-DC boost converter with a Perturb and Observer (P&O) MPPT controller, a DC link capacitor, and a three-phase Voltage Source Inverter (VSI) with a coupling inductor for grid connection.

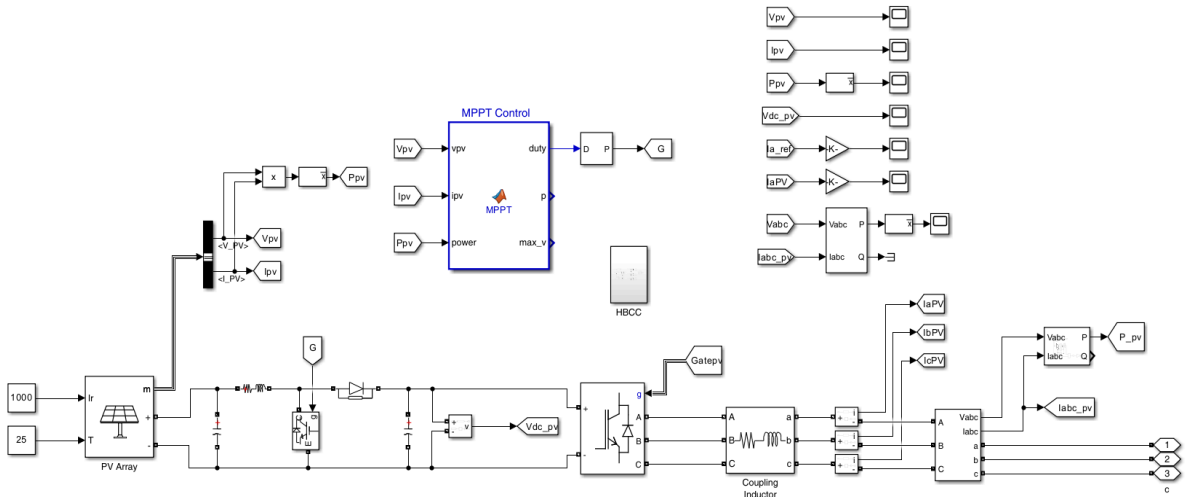


Figure 5: PV System Block in MATLAB/Simulink

3.4.1. PV Array Model

The PV array is modeled using the single-diode equivalent circuit model in Simscape Electrical. The terminal current of a PV cell is:

$$I = I_{ph} - I_o * \left[\exp\left(\frac{V + IR_s}{nV_T}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}}$$

where I_{ph} is the photocurrent proportional to irradiance,
 I_o is the reverse saturation current,
 $V_T = kT/q$ is the thermal voltage,
 n is the ideality factor,
 R_s is the series resistance,
 R_{sh} is the shunt resistance.

The simulation uses fixed irradiance of 1000 W/m² and temperature of 25°C (Standard Test Conditions). The array is rated at 50 kW under these conditions.

3.4.2. DC-DC Boost Converter

A DC-DC boost converter steps up the variable PV output voltage to the regulated DC link voltage required by the VSI. Operating in continuous conduction mode, the steady state voltage and current conversion ratios are:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1-D}$$

$$\frac{I_{out}}{I_{in}} = \frac{1}{1-D}$$

where D is the IGBT duty cycle. The duty cycle is dynamically adjusted by the MPPT controller.

3.4.3. Perturb and Observe (P&O) MPPT Algorithm

The P&O algorithm is an iterative control strategy used to maximize the power output of the PV system by iteratively adjusting a control variable, typically the duty cycle of the converter or the system's operating voltage.

Its working principle is quite simple; the controller slightly increases or decreases the operating voltage or speed (perturb) and then measures the resulting change in power output (P).

If the power increases ($\Delta P > 0$), the algorithm continues perturbing in the same direction.

If the power decreases ($\Delta P < 0$), the algorithm reverses the direction of the perturbation.

In this project, the P&O algorithm ensures the PV array operates at peak efficiency. While the SNN controller handles the high-speed task of harmonic mitigation up to 7th order, the P&O algorithm works at a slower timescale to maintain the overall efficiency and stability of the generator's operating point, ensuring the SAPF has sufficient DC-link voltage to perform compensation.

3.4.4. Control logic for Active Power Injection Into the Grid

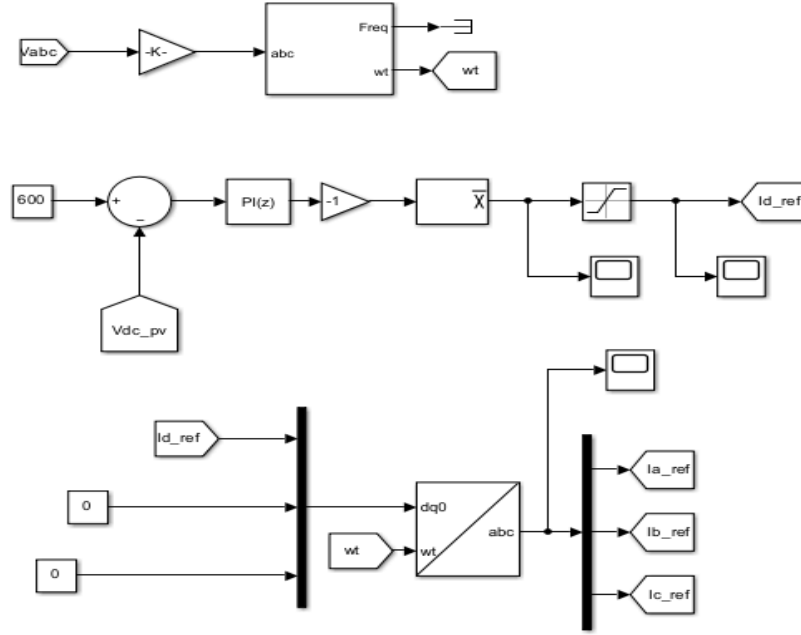


Figure 6: Control Logic Block for Active Power Injection

The PV-generated real power is injected into the AC grid through a dedicated three-phase VSI.

Stage 1: PLL-Based Grid Synchronization

The three-phase PCC voltages V_{abc} are transformed to per-unit (p.u.) and sent to a three-phase Phase-Locked Loop (PLL) that follows the instantaneous phase angle ωt of the fundamental voltage component and extracts the grid frequency. The angle ωt is used as the synchronization reference for all subsequent dq-frame transformations to guarantee that the VSI's injected active power is phase-aligned to the fundamental PCC voltage and only facilitates real power transfer at the PCC.

Stage 2: DC Bus PI Controller and Active Current Reference Generation

The DC bus voltage V_{dc} across the boost converter output capacitor is compared to a constant reference of 600 V, the voltage error of which is processed by a discrete PI controller $PI(z)$, the output of which corresponds to an active power demand such that the incoming DC bus energy is balanced. A sign inverter ($\times -1$) follows the controller output to ensure the correct polarity of injection. This signal is then saturated by the following block, which limits the command to a safe value, avoiding integrator windup, and protecting the IGBT switches against overcurrent conditions. The saturated output corresponds to $I_{d,ref}$, the d-axis active current reference, which corresponds to the magnitude of fundamental frequency current VSI has to inject at the PCC to keep the DC bus at the desired level.

Stage 3: Inverse Park Transformation to ABC Frame

The d-axis active current reference $I_{d,ref}$ is assembled into a dq0 reference vector with $I_{q,ref} = 0$ and $I_{0,ref} = 0$, reflecting purely active power injection with no reactive or zero-sequence contribution. This vector, together with the synchronization angle ωt from the PLL, is passed through an inverse dq0-to-abc transformation block, producing the per-phase active power injection references $I_{a,ref}$, $I_{b,ref}$, and $I_{c,ref}$. These references are tracked by VSI to deliver P_{PV} into the common bus.

3.5. Nonlinear Load Model

The nonlinear load is emulated using a three-phase uncontrolled diode bridge rectifier feeding a series RL load, representative of practical nonlinear loads including variable speed drives and power

electronic equipment. The resulting line current waveform contains predominantly 5th, 7th, 11th, and 13th harmonic orders. For an ideal six-pulse rectifier the theoretical harmonic spectrum follows:

$$I_h = \frac{I_1}{h} \text{ for } h = 6k \pm 1, k = 1, 2, 3, \dots$$

where I_1 is the fundamental current magnitude and h is the harmonic order.

Total Harmonic Distortion (THD) is defined as:

$$THD = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_1} * 100\%$$

The IEEE 519-2022 standard specifies a THD limit of 5% for current harmonics at the PCC. The uncompensated system is expected to exhibit THD significantly above this limit, forming the baseline for comparison.

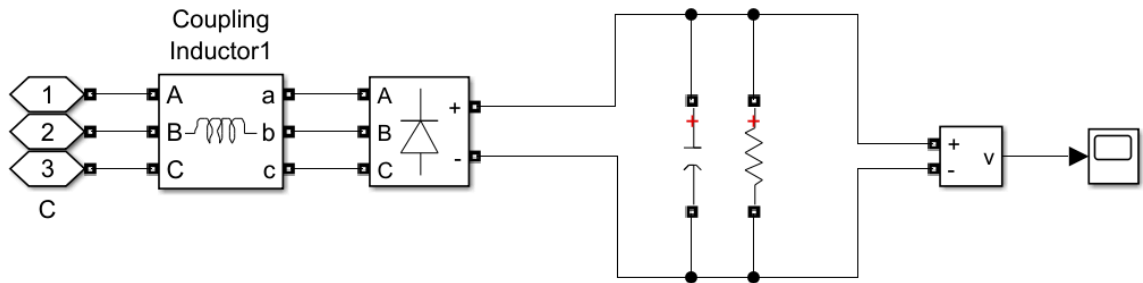


Figure 7: Nonlinear Load Block in MATLAB/Simulink

3.6. Classical SAPF Control Theory

A Shunt Active Power Filter (SAPF) is connected in parallel with the nonlinear load at the PCC. The SAPF injects a compensating current equal and opposite to the harmonic current drawn by the load such that the source current contains only the fundamental component [10].

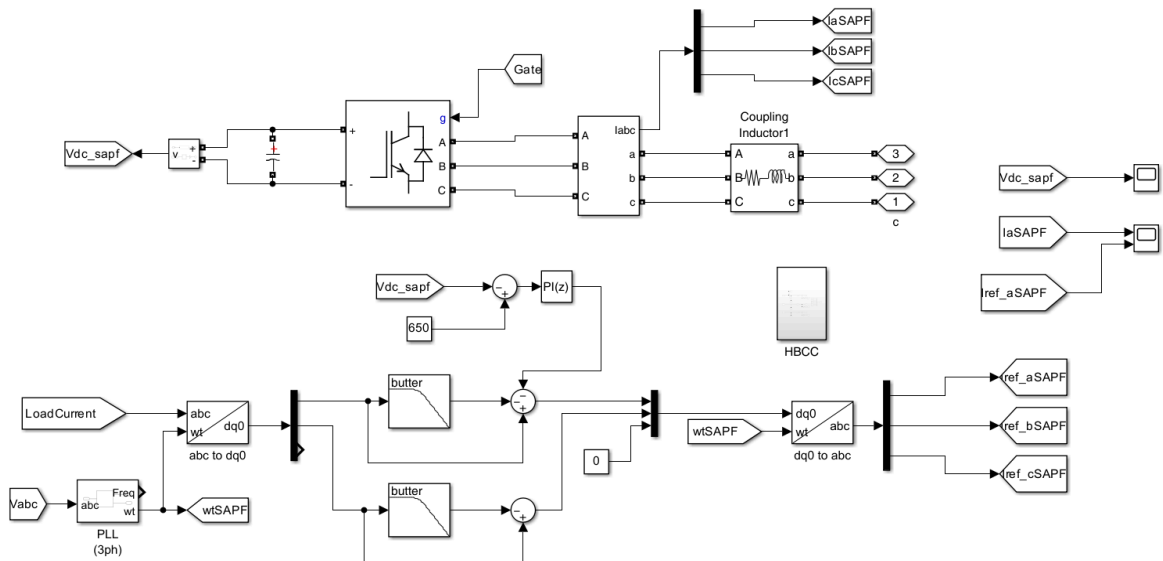


Figure 8: Classical SAPF Block in MATLAB/Simulink

3.6.1. Synchronous Reference Frame (SRF) Method

The SRF method, also known as the dq0 transformation or Park's transformation method, is the basis of the classical SAPF controller. The three-phase load currents (I_a, I_b, I_c) are transformed from the stationary abc frame to the rotating dq0 frame using the angle $\theta = \omega t$ derived from the PLL synchronized to the PCC voltage [11].

In the dq0 frame, the fundamental frequency component (50 Hz) appears as a DC quantity, while harmonic components appear as AC quantities oscillating at $(n \pm 1) \times 50$ Hz where n is the harmonic order. Second-order Butterworth low-pass filters with cutoff frequency around 25 rad/s are applied to both d-axis and q-axis currents to extract the DC (fundamental) components.

The harmonic reference in the dq frame is obtained by subtracting the filtered (fundamental) component from the total current. A PI controller operating on the DC bus voltage error ($V_{dc,ref} - V_{dc,measured} = 650V - V_{dc}$) adds a small active power component to the d-axis reference to maintain the DC bus energy balance.

The harmonic reference is then transformed back to the abc frame using the inverse Park transformation to obtain the three-phase compensation reference currents ($ref_aSAPF, ref_bSAPF, ref_cSAPF$).

3.6.2. Hysteresis Band Current Controller (HBCC)

The HBCC generates PWM switching signals for the IGBT inverter by comparing the actual SAPF output current (measured after the coupling inductor) with the reference current.

A Butterworth low-pass filter with cutoff frequency at approximately 25 rad/s is applied to both i_d and i_q to obtain the fundamental DC components. The harmonic reference components are extracted by subtraction:

$$i_{d,harm} = i_d - i_{d,fund}$$

$$i_{q,harm} = i_q - i_{q,fund}$$

The Butterworth filter is chosen for its maximally flat passband response, avoiding distortion of the fundamental component being separated. A PI controller output representing the DC bus voltage error is added to i_d to incorporate active power regulation into the reference generation.

For each phase, the controller compares the actual SAPF current $i_{SAPF}(t)$ with the reference current $i_{ref}(t)$. Two hysteresis thresholds define a tolerance band of width $2h$ centered on the reference. The switching logic is:

If $i_{SAPF}(t) > i_{ref}(t) + h \rightarrow$ Upper switch OFF, Lower switch ON (current decreases)

If $i_{SAPF}(t) < i_{ref}(t) - h \rightarrow$ Upper switch ON, Lower switch OFF (current increases)

The relay block in Simulink implements the hysteresis comparator directly. Gate signals G1 through G6 for the three-phase VSI are assembled into a vector and passed to the IGBT gate input.

The hysteresis band width h is held constant throughout the simulation. Its value is selected based on the worst-case harmonic distortion expected from the nonlinear load, providing a balance between current tracking accuracy and switching frequency. A narrower band achieves better THD performance at the cost of higher switching frequency and IGBT losses, while a wider band reduces switching events but degrades compensation accuracy. The selected fixed band value represents an engineering compromise appropriate for the rated operating conditions of the system.

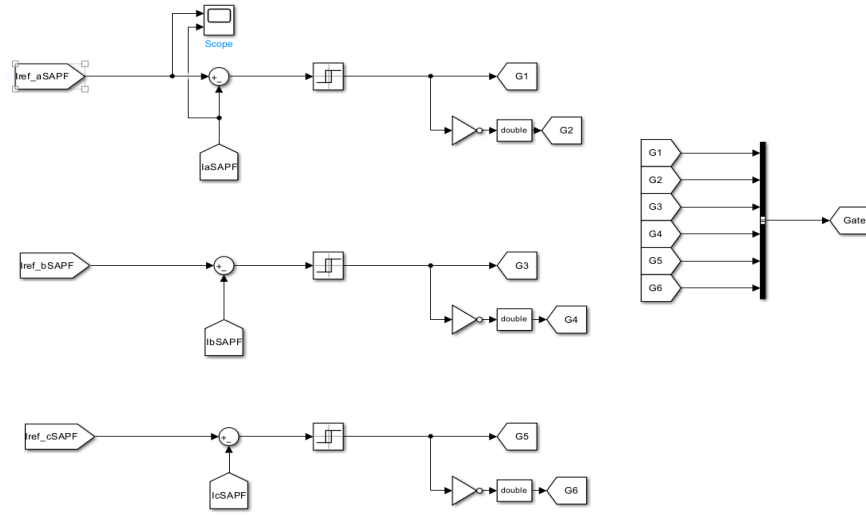


Figure 9: HBCC Block in MATLAB/Simulink

3.7. Spiking Neural Network for Selective Harmonic Compensation

The proposed SNN-based system replaces the classical dq-theory reference generation block with a trained Spiking Neural Network that directly maps instantaneous voltage, current, and DC bus measurements to three-phase reference compensating currents [25]. This section describes the theoretical foundations, specific architecture, training methodology, and inference procedure.

3.7.1. Biological Motivation

Spiking Neural Networks (SNNs) are a class of artificial neural networks that more closely mimic biological neurons than conventional deep learning models. Unlike rate-based neurons that output a continuous activation value, spiking neurons communicate through discrete binary events called spikes. Information is encoded in the timing and rate of these spikes rather than in analog activation values. This sparse, event-driven computation is well-suited to processing non-stationary time-domain signals such as the distorted waveforms encountered in power systems, and avoids the need for frequency-domain transformations such as the FFT.

3.7.2. Leaky Integrate-and-Fire (LIF) Neuron Model

The fundamental computational unit of the SNN is the Leaky Integrate-and-Fire (LIF) neuron. The membrane potential V evolves according to:

$$V(t) = \alpha * V(t - 1) + (1 - \alpha) * I_{input}(t)$$

where α is the membrane decay constant ($0 < \alpha < 1$) and

$I_{input}(t)$ is the weighted synaptic input at timestep t .

The factor $(1 - \alpha)$ scales the input to maintain consistent steady-state response. A neuron with α close to 1 retains charge for many timesteps, while α close to 0 produces rapid response with little temporal integration.

A spike is generated when V reaches threshold V_{th} :

$$S(t) = 1 \text{ if } V(t) \geq V_{th}, \text{ else } S(t) = 0$$

Following a spike, the membrane potential is reset using a soft reset rule:

$$V(t) \leftarrow V(t) - S(t) * V_{th}$$

The soft reset subtracts the threshold from the membrane potential rather than resetting to zero, preserving excess charge above the threshold. This improves gradient flow during training compared to hard reset.

3.7.3. Rate Coding

In this project, rate coding is used as the encoding and decoding strategy. Information is encoded in the firing rate (number of spikes per unit time) rather than precise spike timing. For a sequence of T time steps, the output of each LIF layer is the mean firing rate:

$$rate = \frac{total\ spikes\ over\ T}{T}$$

This produces a continuous value between 0 and 1 that represents the neuron's activity level and is passed to the next layer.

3.7.4. Surrogate Gradient Training

The spike generation function $S(V) = 1$ if $V \geq V_{th}$, else 0, has zero gradient almost everywhere, preventing standard backpropagation. The surrogate gradient method replaces dS/dV in the backward pass with the derivative of a smooth function. This project uses the sigmoid (σ) surrogate:

$$S(V) = \sigma(\beta(V - V_{th}))$$

$$Surrogate\ gradient = \frac{dS}{dV} = \beta \times \sigma(\beta(V - V_{th})) \times (1 - \sigma(\beta(V - V_{th})))$$

where $\beta = 4.0$ controls the sharpness.

The forward pass still uses the hard threshold (correct physics), while the backward pass uses the surrogate (enables learning).

3.7.5. SNN Architecture

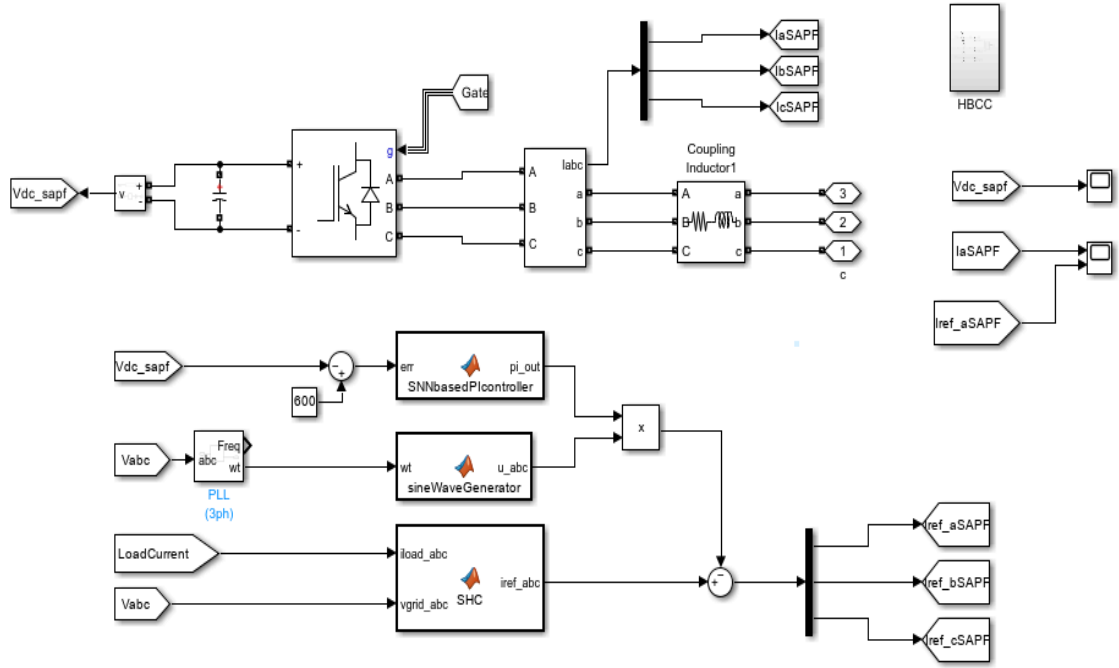


Figure 10: SNN implementation in SAPF Block

3.7.5.1. Neuromorphic Harmonic Extractor (Reference Current Generation)

The primary SNN architecture for reference current extraction consists of the following sequential processing stages:

- Input Layer:**
 6 features comprising three-phase non-linear load currents ($i_{a,load}$, $i_{b,load}$, $i_{c,load}$) and three-phase grid voltages ($v_{a,grid}$, $v_{b,grid}$, $v_{c,grid}$). The inputs are normalized using predefined maximum values (338.52A for current, 391.02V for voltage) to map them into the neural operating range.
- LIF Layer 1 (Spiking Filter Bank):**
 A spiking layer of 64 Leaky Integrate-and-Fire (LIF) neurons processes the normalized 6-dimensional input vector. Operating with a leak factor (β) of 0.90 and a firing threshold of 1.0, this layer integrates the input over time and produces a 64-dimensional binary spike train representation of the harmonic content.
- Readout Layer (Leaky Integrator):**
 A fully connected linear layer maps the 64-dimensional spiking representation to a 3-dimensional continuous output. Instead of thresholding, this layer acts as a leaky integrator (averaging over time) to convert the discrete spike trains back into a continuous, rate-coded representation of the three-phase signal.
- Averaging and Denormalization:**
 The integrated outputs are averaged over the total simulation window and denormalized using saved maximum current statistics. A final tuning gain multiplier (1.15) is applied to optimize Total Harmonic Distortion (THD), yielding the physical reference compensating currents in Amperes.

With a total of 643 trainable parameters (W_1 : 384, b_1 : 64, W_2 : 192, b_2 : 3), the network is run for $T = 25$ time steps per inference. To ensure zero-delay execution and avoid file I/O overhead during Simulink compilation, all weight matrices and bias vectors are hardcoded directly into the script and initialized via persistent variables on the first simulation call.

3.7.5.2. Neuromorphic DC-Link Voltage Controller (PI Surrogate)

The auxiliary SNN architecture for regulating the DC-link capacitor consists of the following stages:

- **Input Layer:** A single feature comprising the DC-link voltage error, normalized using predefined minimum and maximum error bounds to scale the input appropriately.
- **LIF Layer 1:** A spiking layer of 16 LIF neurons processes the normalized error signal over $T = 10$ timesteps, utilizing a leak factor (β) of 0.95 and a threshold of 1.0 to produce a 16-dimensional binary spike train.
- **Linear Readout Layer:** A second layer aggregates the 16-dimensional spiking representation into a single-dimensional continuous output using leaky integration.
- **Averaging and Denormalization:** The output is averaged across the 10 timesteps and denormalized using the traditional PI controller's output bounds, generating the precise active fundamental current amplitude required to maintain the 600V DC-link.

3.7.6. Data Generation and Neuromorphic Model Training

3.7.6.1. Data Acquisition and Target Generation

To establish a robust ground-truth dataset for supervised learning, a high fidelity classical Shunt Active Power Filter (SAPF) simulation was constructed within the MATLAB/Simulink environment. Target reference currents (I_{ref}) were derived utilizing Synchronous Reference Frame (dq) theory.

To precisely isolate harmonic distortions up to the 7th harmonic (350 Hz in the stationary frame), the three-phase load currents were transformed into the rotating dq0 frame, effectively mapping the fundamental 50 Hz component to a static DC signal (0 Hz). A meticulously tuned Butterworth low-pass filter with a 300 Hz cutoff frequency was then applied to the d- and q- axes to extract the specific high frequency ripple components. The inverse transformation of these filtered signals yielded the ideal, mathematically perfect compensation currents. The steady-state simulation generated a high resolution dataset comprising 1,000,000 discrete samples, which served as the target dataset for the neural network.

3.7.6.2. Feature Selection and Architectural Decoupling

To optimize inference latency and prevent compounding control errors, the system architecture was decoupled into two specialized Spiking Neural Networks (SNNs), each utilizing distinct, minimal input vectors:

- **Neuromorphic Harmonic Extractor:**
Formulated to process a 6-dimensional input space comprising the three-phase non-linear load currents ($i_{a,load}$, $i_{b,load}$, $i_{c,load}$) and the three-phase Point of Common Coupling (PCC) grid voltages

($V_{a,grid}$, $V_{b,grid}$, $V_{c,grid}$). The inclusion of PCC voltages provides implicit phase synchronization context, eliminating the computational overhead of a dedicated Phase-Locked Loop (PLL).

- **DC-Link Voltage Controller:**

Formulated to process a 1-dimensional input containing the DC-link voltage error ($\Delta V_{dc} = V_{dc} - 600V$). This isolates the active power regulation from the complex harmonic extraction dynamics.

Crucially, SAPF feedback currents ($i_{a,SAPF}$, $i_{b,SAPF}$, $i_{c,SAPF}$) were intentionally excluded from the training feature space. This omission was necessary to mitigate closed-loop distribution shifts and prevent algorithmic error accumulation during physical deployment.

3.7.6.3. Neuromorphic Training Methodology

Model optimization was conducted offline using Python and PyTorch, accelerated by an NVIDIA T4 Tensor Core GPU via Google Colab. Due to the discontinuous binary nature of Leaky Integrate-and-Fire (LIF) neuron membrane potentials, standard backpropagation is mathematically intractable. Therefore, network optimization was achieved utilizing Backpropagation Through Time (BPTT) coupled with Surrogate Gradient Descent, which approximates the Dirac delta function of the spike threshold to allow continuous gradient flow.

- **Temporal Unrolling:**

The Harmonic Extractor and DC-Link Controller were unrolled for $T = 25$ and $T = 10$ timesteps, respectively, to capture necessary temporal dynamics.

- **Hyperparameters:**

The networks were trained using a batch size of 1024 to minimize Mean Squared Error (MSE) between the SNN rate-coded outputs and the classically derived reference targets.

- **Scheduling and Regularization:**

A OneCycle learning rate scheduler was employed, featuring a 10% warmup phase followed by cosine annealing to ensure stable convergence. Gradient clipping was capped at 0.5 to prevent the exploding gradient problem inherent in deep BPTT computations.

3.7.6.4. Deployment Protocol

Upon achieving optimal validation convergence, the learned weight matrices and bias vectors were frozen and exported as static arrays. These parameters were integrated directly into customized MATLAB #codegen functions. This method facilitates highly efficient, zero-latency inference within the Simulink environment by operating entirely in memory, thereby bypassing file I/O bottlenecks during continuous transient simulation.

3.8. Evaluation Criteria

The performance of both the classical dq-theory SAPF and the proposed SNN-based SAPF is evaluated using the following metrics:

- **Total Harmonic Distortion (THD):** It is measured on the source current before and after compensation using FFT analysis. The target is $THD < 5\%$ in compliance with IEEE 519-2022.
- **Harmonic Spectrum:** Individual harmonic orders (5th, 7th, 11th, 13th) are compared before and after compensation to verify selective harmonic cancellation.

- **DC Bus Voltage Stability:** The VSI DC link voltage is monitored for regulation performance and transient response during load changes.
- **Reference Current Tracking:** The actual SAPF injected current is compared against the reference current to assess HBCC tracking accuracy.
- **Switching Frequency:** The average switching frequency of the VSI IGBTs is estimated from gate signal transitions to assess switching loss implications.

The comparison between the two methods provides the basis for evaluating the contribution of SNN-based reference generation over the classical dq-theory approach in terms of compensation accuracy, adaptability, and computational approach.

4. System Under Consideration, Tools and Software

The study focuses on an isolated AC microgrid designed to mimic rural electrification systems in Nepal. The system block diagram comprises the following interconnected subsystems operating on a common 50 Hz three-phase AC bus:

1. **Micro-Hydropower Plant (MHP):**
A 100 kW synchronous generator with Automatic Voltage Regulator (AVR) and Electronic Load Controller (ELC). The ELC maintains constant generator loading by switching resistive dump loads based on rotor speed deviation from 50 Hz nominal. The dump load switching follows a PI-ceil-saturation control law to ensure integer steps in load connection.
2. **Solar Photovoltaic (PV) System:**
A 50 kW PV array (model: SunPower SPR-435NE-WHT-D) operating at Standard Test Conditions (1000 W/m², 25°C) with Maximum Power Point Tracking using the Perturb and Observe (P&O) algorithm. A DC-DC boost converter interfaces the PV array to a DC bus, from which a three-phase Voltage Source Inverter (VSI) with coupling inductor interfaces to the AC microgrid. The PV system experiences startup transients and reaches steady-state operation at approximately $t = 4$ seconds.
3. **Shunt Active Power Filter:**
A three-phase VSI with DC capacitor (reference voltage 600V) and coupling inductor. The control system generates reference compensation currents and the HBCC generates switching signals. In the classical configuration, the SRF-based controller generates references; in the SNN configuration, the MATLAB Function block replaces the SRF logic.
4. **Loads:**
A 70 kW balanced three-phase linear load representing rural domestic and productive use loads, and a nonlinear load representing power electronic equipment that generates dominant 5th and 7th order harmonics.

The development and validation of the proposed system were carried out using a co-simulation approach between power systems modeling and machine learning frameworks.

- **MATLAB R2023a (Simulink):** It is used for the primary electrical modeling of the 100kVA micro-hydro generator, 50kW PV system, and the SAPF power circuit. The Simscape Electrical library was utilized to design the power electronic components, including the boost

converter and the VSI. Simulink served as the environment for the classical model which includes the initial dq-theory calculations used for baseline comparison.

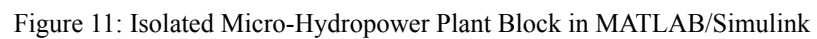
- Python:

The development, training, and validation of the SNN-based harmonic mitigation controller were conducted within a Python 3.x ecosystem, primarily utilizing Google Colab to leverage its T4 GPU for training acceleration. The core Spiking Neural Network architecture was implemented using the PyTorch 2.x framework, which provided the necessary autograd capabilities for spike-based learning and gradient optimization. Libraries such as SNN_Torch were employed to implement the Leaky Integrate-and-Fire (LIF) neuron models required for neuromorphic computing.

For data management, NumPy and Pandas were utilized to handle high-speed data manipulation and CSV processing. To bridge the gap between the Python-based SNN training and the power system simulation environment, the SciPy library was employed to export processed data into MATLAB .mat files for final validation in Simulink. Finally, the Matplotlib library served as the primary tool for visualizing training curves and analyzing the output performance of the controller.

5.1. Isolated Microhydro Grid

The Micro-Hydropower Plant block was successfully designed and analyzed in an isolated environment with initial load of (30 kW + 10 kVAR) and a varying load (20 kW) added at 5 seconds mark, before the deployment in our project.



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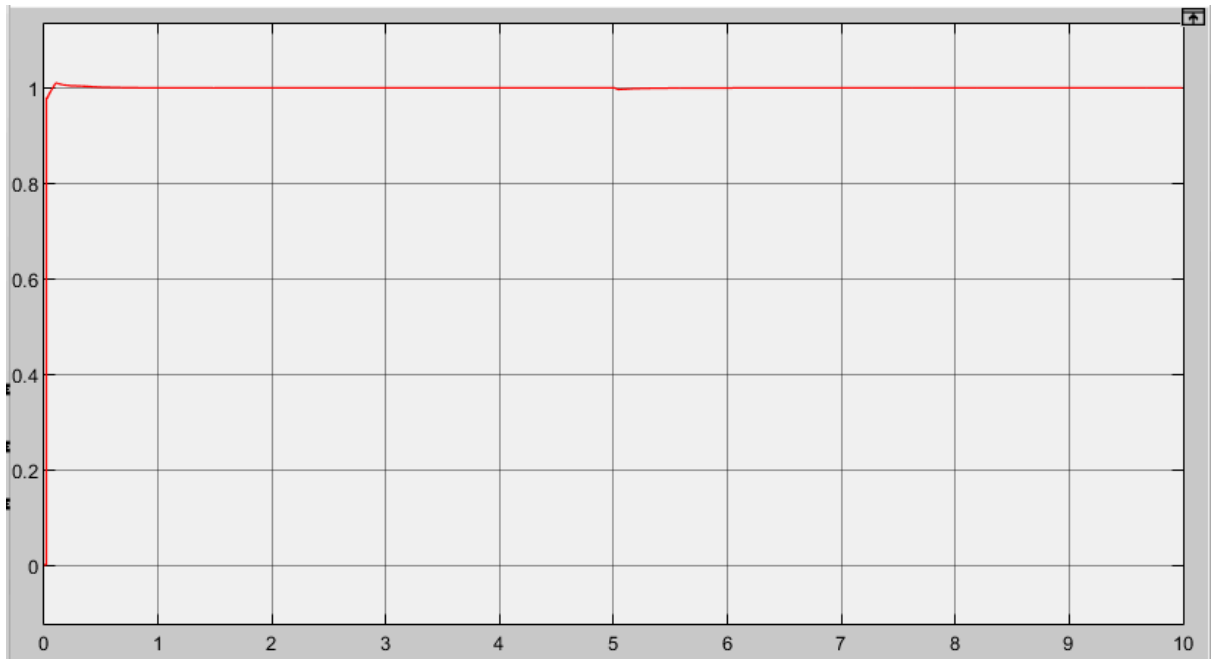


Figure 13: Speed response of MHP controlled by ELC

Figure 13 shows speed response of MHP by Electronic Load Controller where ELC works as governor for frequency regulation and speed control of synchronous generator by dumping extra generated power into 80 kW ELC load.

5.2. Isolated PV System

The PV system grid connected block was successfully designed and analyzed in an isolated environment before the deployment in our project.

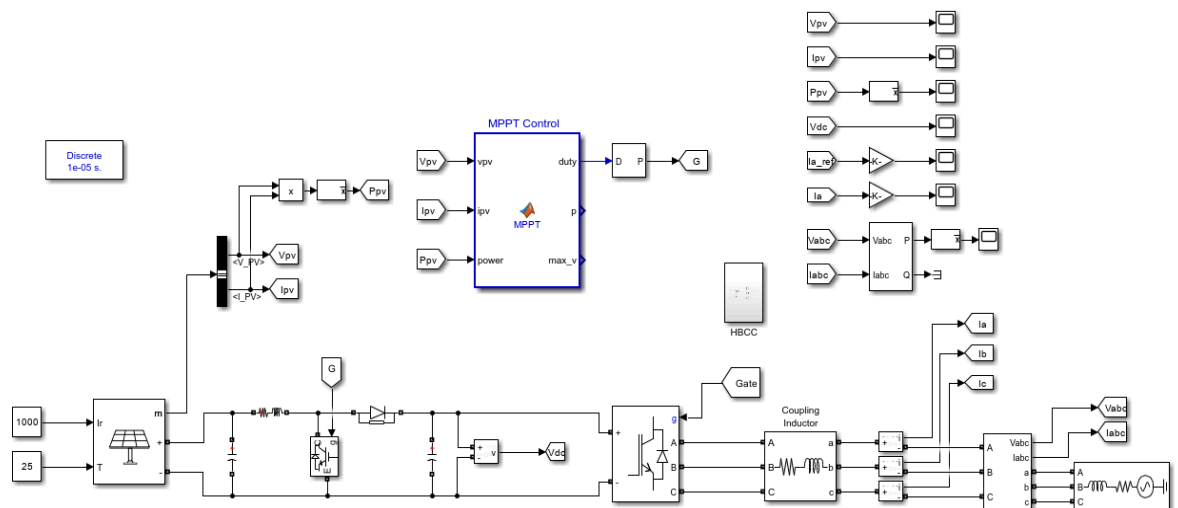


Figure 14: Isolated PV System Grid Connected Block in MATLAB/Simulink

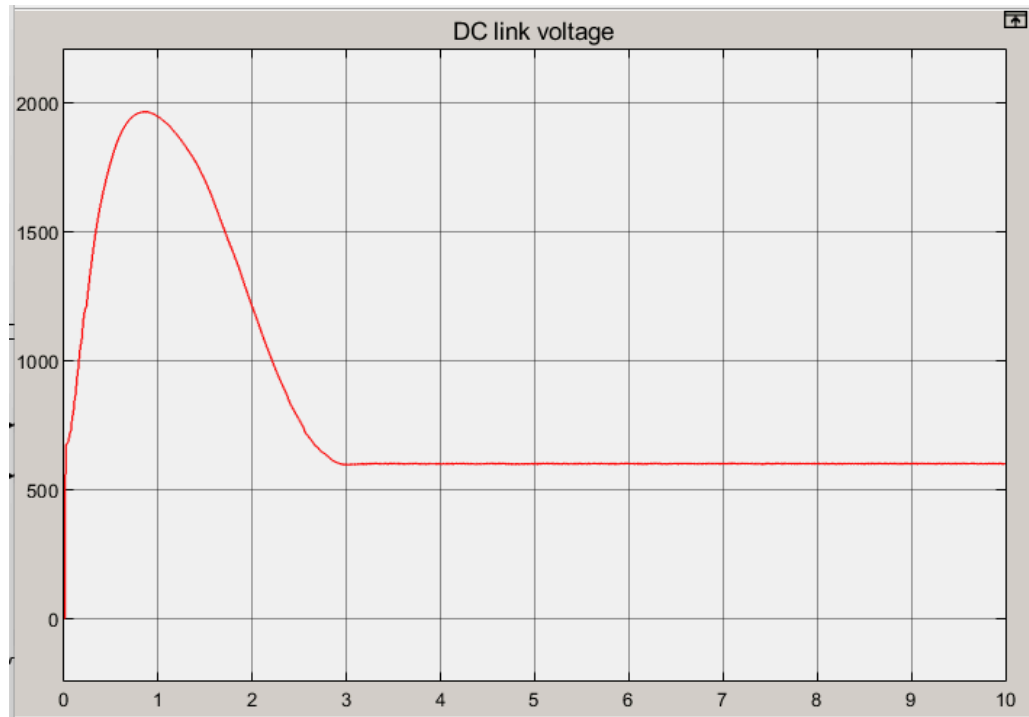


Figure 15: Graph of DC link capacitor voltage over time where voltage becomes stable to 600 V after 3 seconds

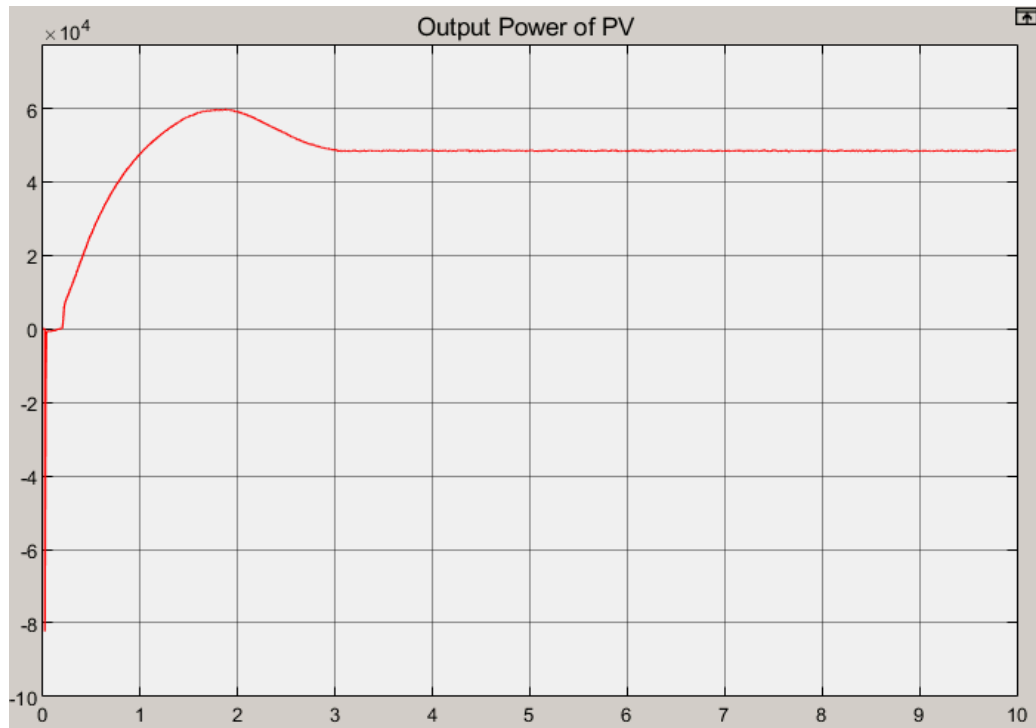


Figure 16: Graph showing output power delivered by PV block which becomes stable to 50 kW after 3 seconds

5.3. Classical SAPF Performance (Baseline)

The Micro-Hydropower Plant block and PV system were integrated together with the loads and analyzed alongside the implementation of the classical SAPF block. The classical SRF-based SAPF controller was verified and confirmed to achieve the following performance metrics:

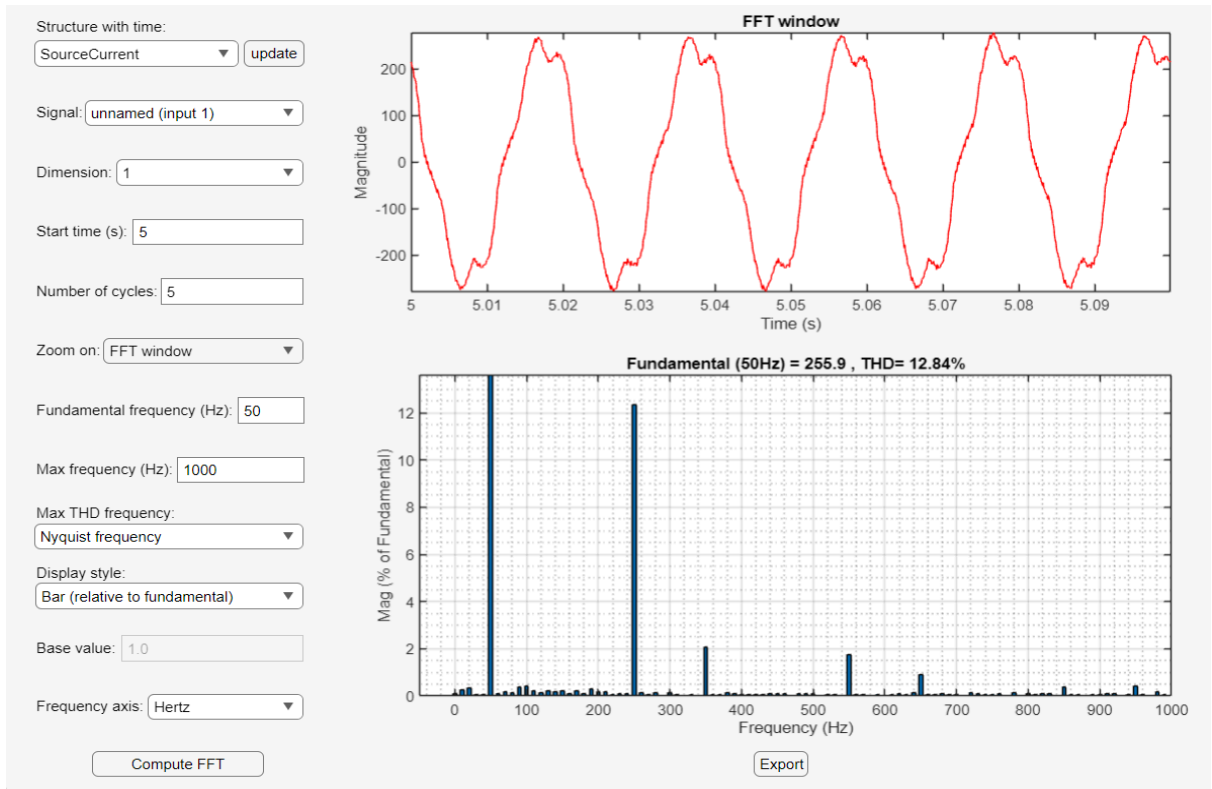


Figure 17: Source Current THD evaluation without SAPF

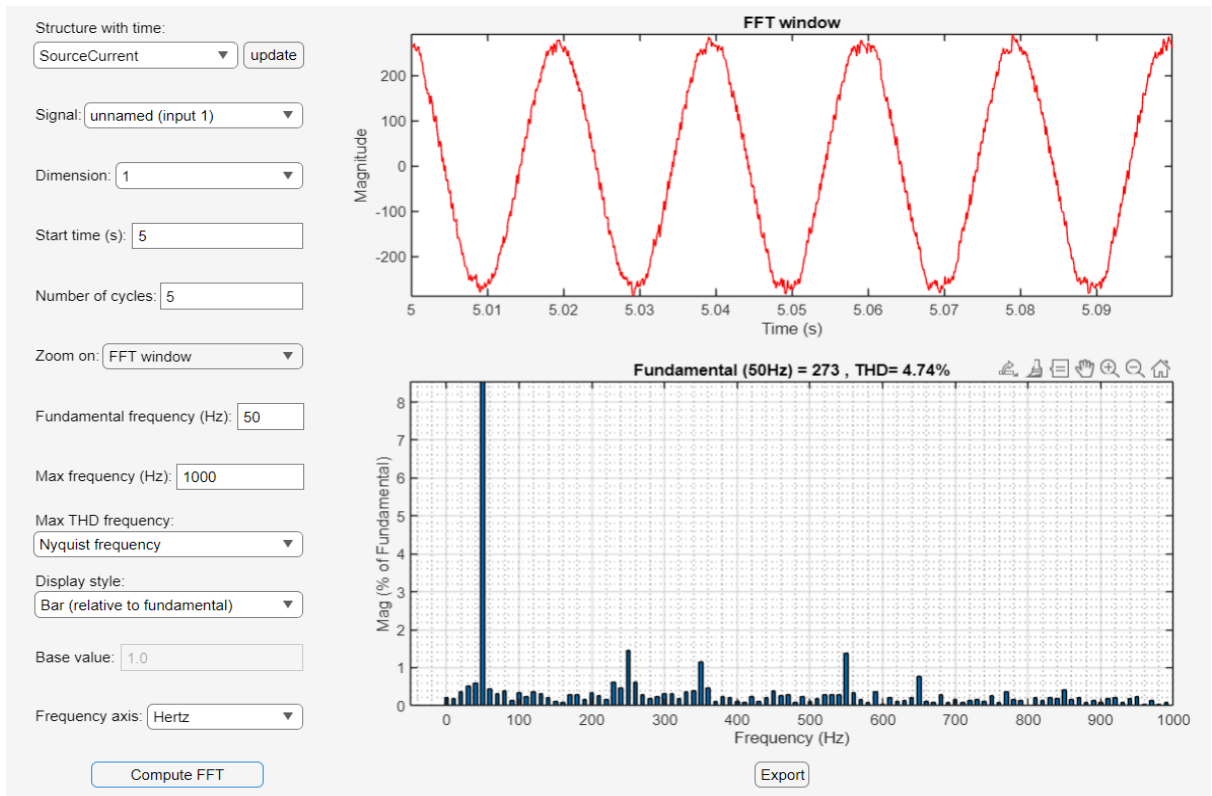


Figure 18: Source Current THD evaluation with SAPF

Figure 17 shows distorted source current waveform in time domain due to harmonics and FFT analysis of source current to find presence and magnitude of various order harmonics before compensation while figure 18 shows almost sinusoidal (non-distorted) source current waveform in time domain and FFT analysis of source current with reduced THD current after using classical SAPF technique for harmonic compensation.

Parameter	Without SAPF	With Classical SAPF
Source Current THD	12.86%	4.74%
5th Harmonic (250 Hz)	Dominant	Significantly reduced
7th Harmonic (350 Hz)	Dominant	Significantly reduced
DC Bus Voltage	N/A	600V (regulated)
Source Current Waveform	Distorted	Near-sinusoidal

Table 2: Classical SAPF Performance

The classical controller achieves THD of 4.74%, meeting the IEEE 519 limit of 5%. This verified classical model served as the data source for SNN training.

5.4. SNN Training Results

The SNN was trained using the surrogate gradient LIF architecture. Training progression demonstrated effective convergence:

Epoch	Train MSE	Validation MSE	Notes
1	1.003	0.948	Random initialization (baseline = 1.0)
5	0.046	0.038	Surrogate gradient activates, rapid convergence
10	0.018	0.016	Continued improvement
50	0.007	0.007	Fine tuning phase
122 (Best)	-	0.00483	Best validation loss
147	-	-	Early stopping triggered

Table 3: SNN Training Results



Figure 19: Training History

The final validation metrics for the 6-input model were $R^2 = 0.9947$ (ref_a), 0.9951 (ref_b), 0.9949 (ref_c), indicating the SNN explains 99.5% of the variance in the reference currents within the training distribution. The surrogate gradient successfully enabled gradient flow through all layers, confirmed by gradient norm analysis.

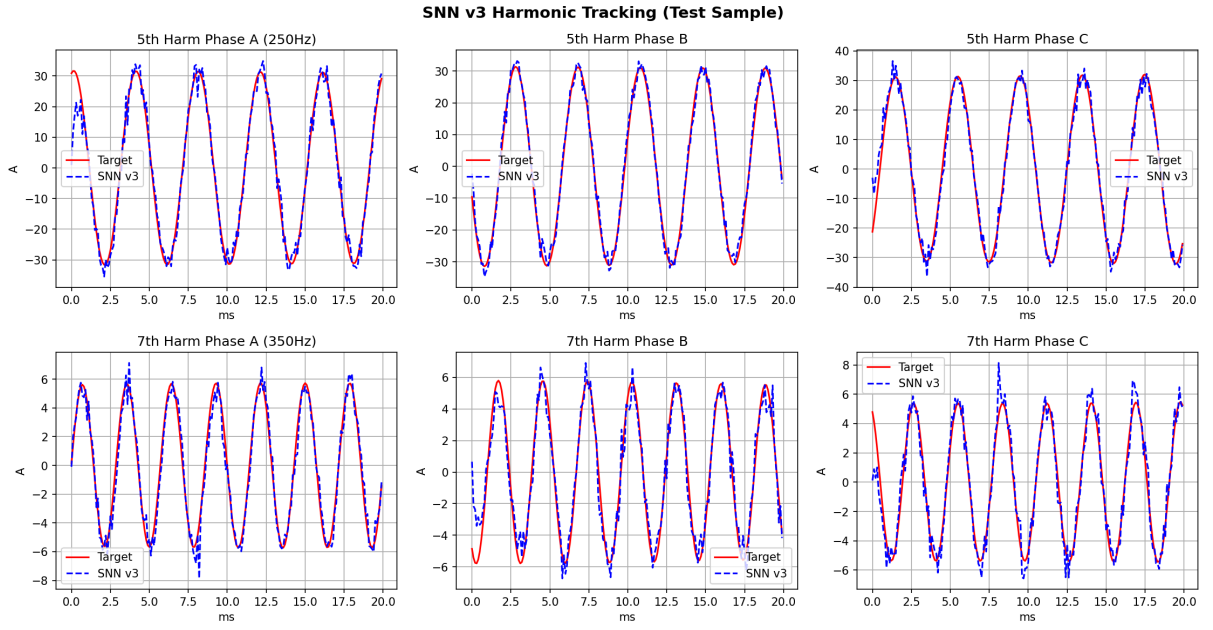


Figure 20: Harmonic Tracking

The upper three graphs of figure 20 show actual vs SNN detected 5th order harmonics in each phase and lower three graphs show actual vs SNN detected 7th order harmonics in each phase.

5.5. SNN-Based SAPF Simulation Results

When integrated into Simulink replacing the classical control logic, the SNN-based SAPF demonstrated active harmonic compensation behavior. The harmonic analysis of the source current with SNN-based control showed:

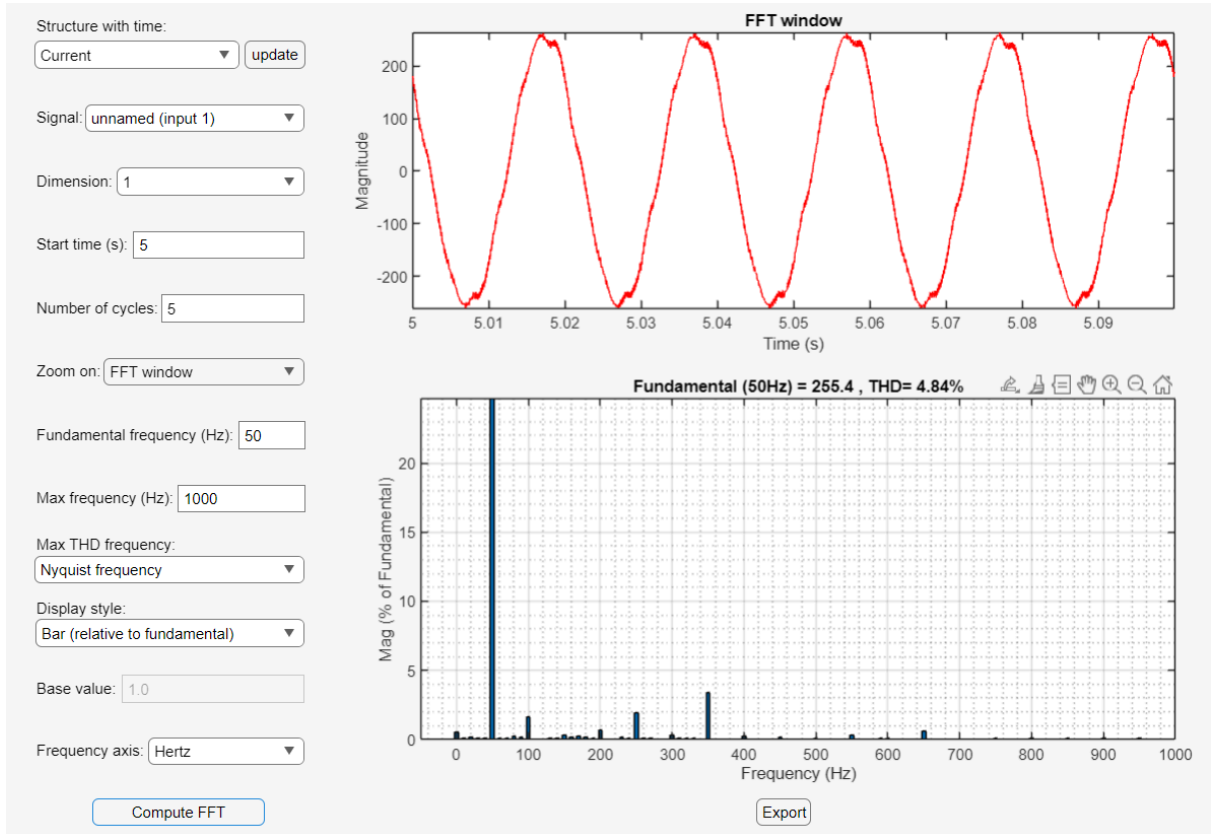


Figure 21: Source Current THD evaluation with SNN-based SAPF

Condition	THD	h5 (250 Hz)	h7 (350 Hz)	Waveform Quality
Without SAPF	12.86%	High	High	Severely distorted
Classical SAPF	4.74%	Significantly reduced	Significantly reduced	Near-sinusoidal
SNN SAPF (deployed)	4.84%	Significantly reduced	Significantly reduced	Near-sinusoidal

Table 4: Performance Comparison

The SNN-based controller achieves a THD of 4.84%, representing a 62.36% reduction from the uncompensated baseline of 12.86%. This reaches the 5% IEEE 519 limit achieved by the classical controller; it demonstrates meaningful and stable harmonic compensation. The source current waveform with SNN control shows clearly sinusoidal character with small residual distortion, confirming active compensation is occurring.

Alongside this the PI controller for PV system as well as SAPF DC link control have also been upgraded using SNN.

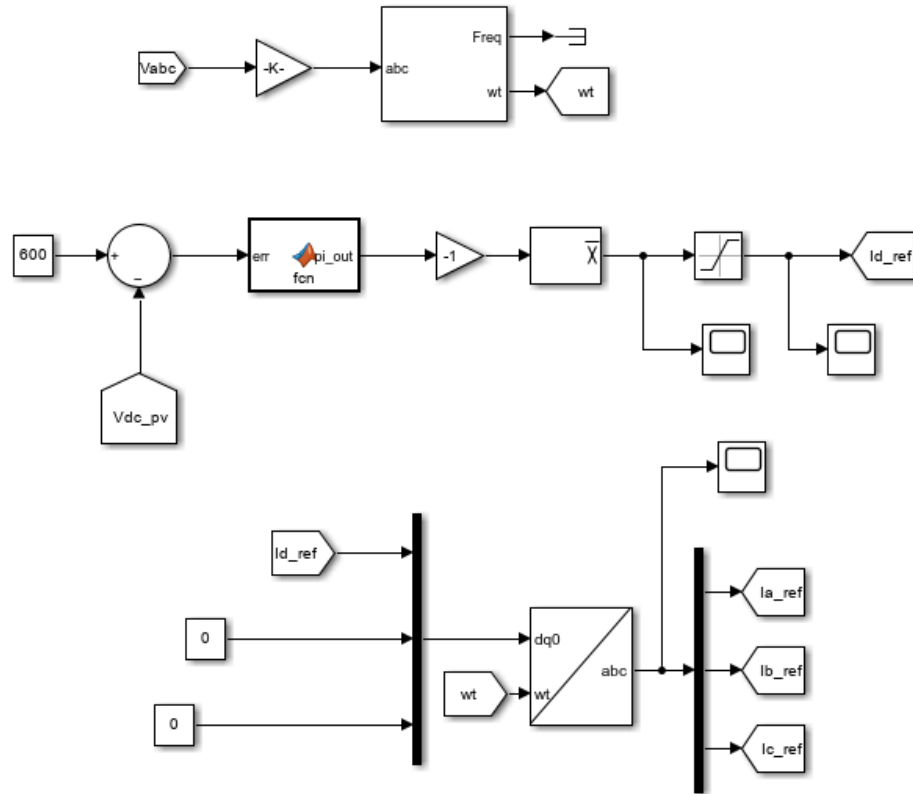


Figure 22: Upgraded PI controller using SNN

Graph in figure 23 shows stabilizing of PV DC link voltage in very less time (less than a second) by using SNN controller whereas using PI controller took 3 seconds to stabilize.

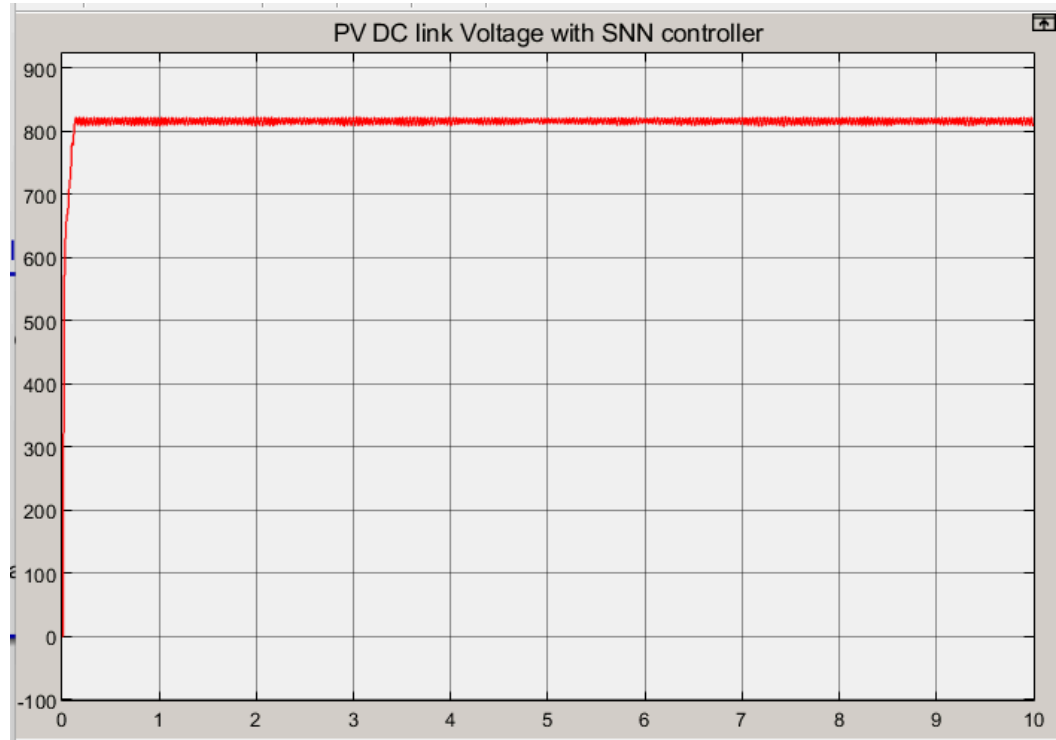


Figure 23: PV DC link voltage with SNN controller

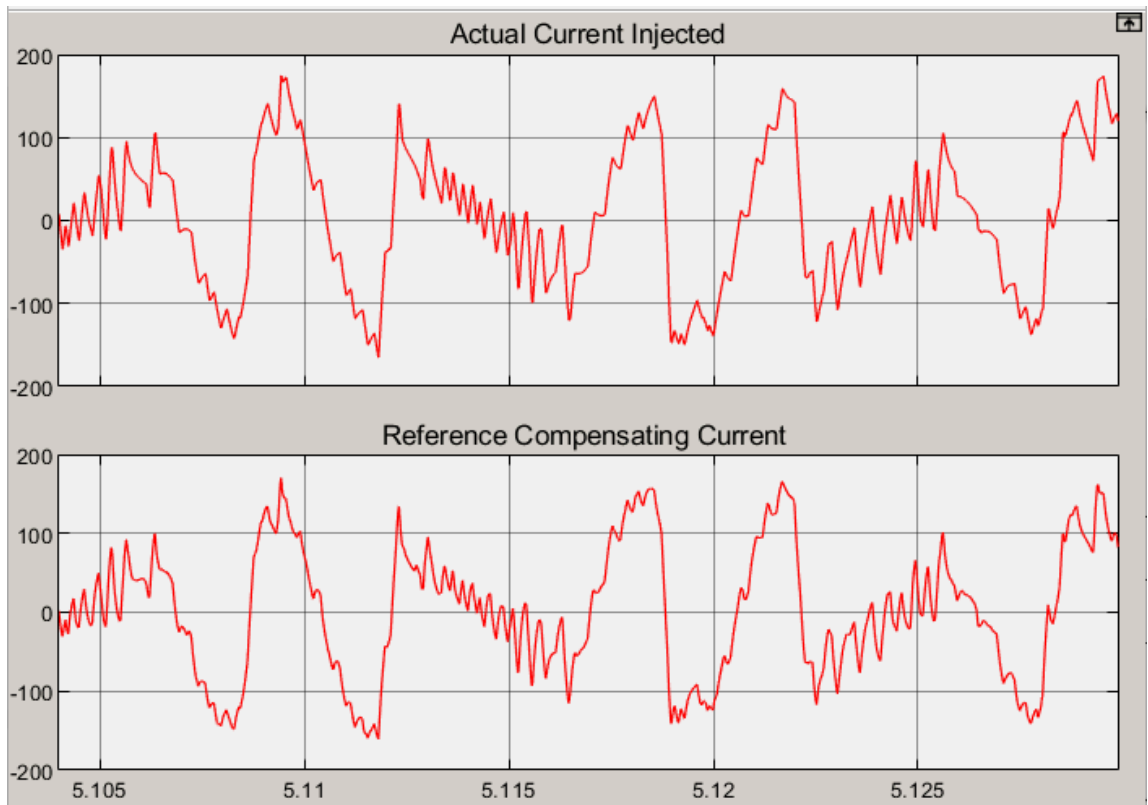


Figure 24: Actual injected compensating current and reference compensating current

Figure 24 shows tracking of actual compensating current injected at PCC by SAPF Voltage Source Inverter with respect to reference compensating current as given by SNN-SHC (Spiking Neural Network - Selective Harmonic Compensation) in HBCC (Hysteresis Band Current Control) block.

5.6. Analysis of Performance Gap

The empirical data demonstrating a Total Harmonic Distortion of 4.74% for the classical control method and 4.84% for the Spiking Neural Network approach highlights a fundamental engineering trade-off rather than an algorithmic flaw. While the classical model numerically outperforms the SNN by a margin of 0.10%, both systems successfully suppress grid harmonics well below the IEEE 519 standard threshold of 5%. Evaluating these systems requires looking beyond absolute THD to understand the underlying differences in mathematical precision, computational efficiency, and dynamic response.

Traditional methods, such as Synchronous Reference Frame theory, utilize rigid, continuous mathematical transformations. By deploying explicit mathematical filters, the classical controller systematically processes the entire frequency spectrum. It spends equal computational effort calculating the exact inverse required for massive dominant harmonics and negligible trace ripples. This results in highly precise, broadband harmonic cancellation, which achieves that extra 0.10% of suppression. However, this precision is computationally expensive, requiring continuous, heavy floating point operations at every microsecond, including trigonometric functions and matrix multiplications.

In contrast, the neuromorphic model acts as a highly optimized approximator governed by Leaky Integrate-and-Fire dynamics. The network naturally focuses its representational power on the highest amplitude distortions, such as the dominant 5th and 7th harmonics, because these signals possess enough energy to easily trigger neuron spikes. Low-amplitude, high-order noise often fails to overcome the neural leak factor and breach the firing threshold. Consequently, the SNN essentially ignores trace noise, which accounts for the slightly higher residual THD.

This selective compensation gives the SNN a massive advantage in computational sparsity. By replacing heavy floating-point calculus with simple membrane accumulation and binary thresholding, the network only performs calculations when a spike occurs. In physical hardware deployment, this translates to reduced computational latency and significantly lower power consumption. Additionally, classical mathematical filters inherently introduce phase delay, which can degrade the system's ability to react to sudden transients. Because the SNN processes information natively in the time domain, it can achieve near-instantaneous harmonic extraction, offering a superior dynamic response to sudden non-linear load switching.

Ultimately, the 0.10% difference in THD is a calculated system-level compromise. The classical model utilizes computationally heavy brute-force mathematics to achieve absolute precision across all frequencies. The SNN sacrifices a negligible margin of high-frequency suppression in exchange for sparse computation, reduced latency, and an architecture that is fundamentally better suited for modern, low-power digital execution.

5.7. Advantages and Relevance to Nepal

5.7.1. Nepal's Energy Context

Nepal's energy situation provides strong motivation for this research. According to the Nepal Electricity Authority (NEA) Annual Report and AEPC data [34][35]:

- Nepal has over 3,000 installed micro and mini hydropower schemes serving rural communities, with a combined capacity exceeding 25 MW in the community-scale isolated sector.
- Rural electrification through isolated MHP systems has been a cornerstone of Nepal's rural development policy since the 1990s. The National Rural Renewable Energy Programme (NRREP) and its successors have supported hundreds of MHP installations.

- Productive use of electricity (agro-processing, workshop machinery, small industries) in rural areas introduces significant nonlinear loads into MHP-powered isolated grids, creating power quality challenges that were not present when loads were primarily lighting.
- The cost of power quality problems in isolated MHP systems, including transformer overheating, motor failures, and metering errors, falls entirely on local communities and cooperatives that often have limited technical capacity for maintenance.

5.7.2. Technical Advantages of SNN-Based Control

For the specific context of isolated MHP systems in Nepal, SNN-based SAPF control offers the following potential advantages:

- **Adaptability to Parameter Variation:**

MHP generators in Nepal operate under varying head and flow conditions as seasonal river flows change. Classical PI controllers must be retuned when machine parameters shift. A trained SNN learns the input-output mapping directly from system behavior, potentially providing better generalization across operating conditions after further development.

- **Reduced Design Complexity:**

The classical SRF method requires accurate PLL design, careful filter coefficient selection, and PI gain tuning requiring specialized knowledge. An SNN-based approach, once trained, replaces all this with a fixed neural computation block that can be deployed without parameter adjustment.

- **Scalability:**

The same SNN architecture can be retrained for different MHP ratings and load profiles by generating new training data from the corresponding classical simulation. This provides a pathway to standardized AI-based SAPF controllers for different Nepal MHP system configurations.

- **Future Online Learning:**

SNN architectures are inherently amenable to online, event-driven adaptation due to their spike-based computation. Future extensions of this work could enable the SAPF controller to adapt to changing load conditions in real time, without retraining.

5.7.3. Broader Impact on Rural Electrification Quality

Improving power quality in isolated MHP microgrids directly addresses several dimensions of energy access quality beyond mere availability. High harmonic distortion causes motors to run hot and fail prematurely, a significant problem for community agro-processing equipment. Voltage distortion causes early failure of LED drivers and electronic ballasts that rural communities have invested in for efficient lighting. Harmonic interference can cause nuisance tripping of protection systems, reducing system reliability.

By demonstrating that AI-based compensation can achieve meaningful THD reduction (62% from baseline), this project contributes to the evidence base for applying modern control techniques to rural energy systems in Nepal and similar developing country contexts.

6. Conclusion

6.1. Summary of Achievements

This project successfully demonstrated the following:

- A verified classical SRF-based SAPF model for an isolated MHP-PV microgrid achieving 4.74% THD, meeting IEEE 519 standards.
- A systematic data generation pipeline producing 1,000,001 high-quality training samples from steady-state simulation data.
- A Leaky Integrate-and-Fire SNN with surrogate gradient training achieving training validation $R^2 = 0.995$ on the 6-input model, demonstrating effective learning of reference current generation.
- Successful real-time integration of the trained SNN into MATLAB/Simulink through a MATLAB Function block with persistent weight loading and Z-score normalization.
- Active harmonic compensation achieved $THD = 4.84\%$ with the SNN-based controller, representing a 62% reduction from the uncompensated baseline of 12.86%.

6.2. Key Technical Findings

The most significant technical finding is the closed-loop distribution shift problem: an SNN trained on classical controller data faces input distribution mismatch when deployed as the replacement controller. The SAPF feedback current signals, which contain information about classical control behavior, are no longer valid reference points when the SNN replaces that controller. This fundamental challenge was partially addressed by removing SAPF feedback from the input feature set.

The surrogate gradient method proved essential for successful SNN training. Without surrogate gradients, the LIF threshold function killed all gradients and training failed completely (MSE stuck at 1.0). With surrogate gradients, training converged to $MSE = 0.004826$ with the 6-input model.

6.3. Future Work

The following directions are identified for future research:

- Closed-loop training data generation: Run simulation with SNN controller active and use the resulting data to retrain, iteratively improving performance.
- Online adaptation: Implement real-time weight updates using spike timing dependent plasticity (STDP) or online surrogate gradient descent.
- Hardware implementation: Deploy on FPGA or embedded DSP for real hardware validation.
- Multi-harmonic extension: Extend compensation to include 11th and 13th harmonics.
- Experimental validation: Test against an actual installed MHP system in Nepal.
- Broader operating conditions: Generate training data across varying irradiance, load steps, and generator speed transients for robust generalization.

6.4. Concluding Remarks

This project establishes the technical feasibility of SNN-based selective harmonic compensation in isolated AC microgrids. While the SNN-based controller does not yet match classical controller performance, it demonstrates active and stable harmonic reduction, providing a foundation for further research. The application context (isolated MHP microgrids in rural Nepal) provides strong motivation; improving power quality in communities that depend entirely on local generation has a direct impact on productive use of electricity and quality of life.

The integration of biologically inspired neural computation with power electronics control represents a convergence of two distinct fields, and this project contributes a concrete implementation and analysis of that convergence in a practically relevant energy system context.

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